

Computer-assisted certification of first-passage modes, unimodality, the closed-form mode formula and two constants for the lazy walk in a reflecting box

Xiaoxiao Zhouyi
Companion note R5

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Note. This document is the companion note R5 of the article “Mean and most probable first-passage times of lazy random walks in reflecting hypercubic lattices” ([paper.pdf](#), Section 7; Supplementary Material, Section S8), in the repository <https://github.com/zhouyi-xiaoxiao/master-dissertation>. In the repository, its scripts are in `code/msc_rigorous_certified/` (the C engine and its helpers in `code/msc_rigorous_certified_c128/`) and its certificates in `data/msc_rigorous_certified/`; paths written `scripts/...`, `scripts/c128/...` and `certs/...` below refer to these folders; the paths in the `% src` comments of the L^AT_EX source are relative to the root of the repository, and the run logs they cite are in `data/msc_rigorous_*/logs/`. Other run logs, the code of the internal checks and working notes mentioned below are not part of the repository; `notes/VERIFICATION.md` summarises how this note was checked.

1 Summary and trusted base

This note replaces “exact to round-off” statements about the first-passage probability mass function (PMF) of the lazy walk in the box $\{0, \dots, N-1\}^d$ by statements that are proved, for finite ranges of N , by integer and interval computations. Every claim is labelled as one of: *Theorem/Lemma/Proposition* (complete proof in this note, standard results cited by theorem number); *computer-assisted Theorem* (the proof is a finite computation in integer or interval arithmetic, together with lemmas proved here that turn the finite computation into the stated claim); *Conjecture*; *Numerical observation*.

The mathematical content consists of the lemmas of Sections 2–3, which reduce each claim about the PMF on the *whole* time axis to finitely many integer inequalities (the key one is the tail lemma, Lemma 5); of an elementary enclosure lemma for the mean first-passage time (Lemma 19); and of results that need no computer at all (Theorem 28, Proposition 32, Lemma 34) or only the interval evaluation of explicit expressions (Theorems 23 and 25).

Results.

- **Theorem 12** (computer-assisted). For $q = 4/5$ and $q = 1/2$, $d = 1, 2, 3$ and every N in the ranges of Table 1 (for $q = 4/5$: $2 \leq N \leq 600$ in one dimension, $2 \leq N \leq 160$ in two, $2 \leq N \leq 60$ in three, and a few larger values): the exact integer mode t^* , uniqueness of the maximiser, strict unimodality on the whole time axis, and an explicit geometric bound for the tail. There are exactly three exceptional cases, all in one dimension and with $N \leq 4$, in which two consecutive values of f coincide exactly.
- **Theorem 13** (computer-assisted). The same analysis for $q = 1$, where the PMF is *not* unimodal in general: the certificates prove multimodality where it occurs.

- **Theorem 21** (computer-assisted). Explicit bounds ε_d for the relative error of the closed form $t_{\text{cf}} = \mathbb{E}T \cdot \ln(2dX)/(X + 2d - 1)$ against the exact integer mode, for every $N \geq 10$ in the certified ranges: $\varepsilon_2 = 0.00970$ and $\varepsilon_3 = 0.00275$ at $q = 4/5$. In one dimension the closed form is certified to be *off by more than 15%*.
- **Theorem 23** (proved; interval enclosure). The one-dimensional continuum constant: the theta series φ has a unique critical point τ^* , a global maximum, and $2\tau^* = 0.33328426549494789874852691\dots$
- **Theorem 25** (interval enclosure) and **Theorem 26** (computer-assisted). The value of $C_2 = 0.15463778781891726249\dots$ and a two-sided bound on the remainder of the two-term MFPT law in two dimensions for $2 \leq N \leq 301$ (self-contained), together with strict monotonicity of the remainder there; with the single-sum formula proved in the companion note, r_N stays below $0.5321405 < C_0$ and strictly increasing for all $N \leq 1500$ (Proposition 27, which also certifies the comparison of the two values of r_N at the junction of the two ranges).
- **Theorem 28** (proved, all N , no computer). In one dimension and for $q \leq 1/2$ the PMF is log-concave, hence unimodal, for every $N \geq 2$. The plateau allowed by the statement does occur ($q = 1/2$, $N = 3, 4$).
- **Theorem 30** (computer-assisted). An exact rounding formula for the one-dimensional mode in the certified range.
- **Proposition 32** (proved, all $N \geq 4$, no computer). In one dimension $f(N - 1) > f(N) < f(N + 1)$ whenever $q > (2N - 4)/(2N - 3)$, so the PMF is *not* unimodal; in particular for $q = 1$ and every $N \geq 4$.
- **Theorems 39–45** (computer-assisted, second trusted base; Section 8). The same statements on much larger ranges, obtained with a second engine written in C: for $q = 4/5$, every $2 \leq N \leq 1500$ in one dimension and $N \in \{2000\}$, every $2 \leq N \leq 301$ in two and $N \in \{350, 401, 450, 500, 600\}$, every $2 \leq N \leq 120$ in three and $N \in \{130, 140, 150, 160\}$; with $\varepsilon_2 = 0.00970$ for $10 \leq N \leq 301$ and $\varepsilon_3 = 0.00275$ for $10 \leq N \leq 120$. These ranges contain every corner-to-corner case of the computations of Sections 5–6 of the article with $q \in \{4/5, 1/2, 1\}$. The C engine reproduces all 1539 verdicts of Theorems 12 and 13. For $q = 1$ it adds four non-unimodal two-dimensional boxes ($N = 86, 92, 136, 149$) to the list of Theorem 13.

What is not proved here. Nothing is claimed for N outside the stated finite ranges, except Theorems 23, 25, 28, Propositions 6 and 32 and Lemma 34. In particular the asymptotic laws (mode $\sim (4/\pi^2 q)N^2 \ln \ln N$ in two dimensions, etc.), unimodality for all N in $d \geq 2$, and the convergence of the lattice ratio mode/MFPT to $2\tau^*$ in one dimension remain outside this note (Section 9).

Trusted base.

- (T1) Python integers (arbitrary precision; operations $+$, $-$, $\times$, floor division, shifts, bitwise and, comparisons, conversion to and from byte strings). The verifying program `scripts/verify_packed.py` uses nothing else: no numpy, no floating point. The same holds for `scripts/verify_exact.py` (`fractions.Fraction`, which is built on the same integers) and for the reference program `scripts/fpcore.py` (which uses numpy only as a container of Python integers).
- (T2) For the generator `scripts/fplimb.py`: numpy `int64` elementwise addition, multiplication, shifts, bitwise and, and `divmod` on non-negative operands, together with the overflow bound of Lemma 8. *No statement of Theorems 12 and 13 rests on (T2)*: every one of the 1539 certificate lines was confirmed by `verify_packed.py`, a program with a different data layout

and a different rounding scheme (Table 1, Section 10). (T2) matters only for anyone who wishes to regenerate the certificates quickly.

(T3) Arb ball arithmetic (FLINT, through `python-flint` 0.9) for π , γ , $\Gamma(1/4)$, $\exp$, $\ln$, $\cos$, with `mpmath.iv` interval arithmetic as a second, separately developed implementation; every sign decision and every enclosure used in a theorem was required to agree between the two.

(T4) Floating-point arithmetic is used only to *guess* (the approximate solution of the MFPT linear system). No statement depends on its accuracy.

(T5) For the extended ranges only (Section 8, Theorems 39–45): C99 unsigned integer arithmetic (`uint64_t`, `unsigned __int128`) as compiled by Apple clang 21 for arm64, in the 366-line program `scripts/c128/fp128.c`; the verdicts are formed from its output with Python integers. The statements that rest on (T5) are labelled as such; Theorems 12 and 13 do not depend on it. Evidence for the correctness of the compiled program (byte-identical output of a Python-integer mirror on 286 small and 10 medium cases, identical verdicts on all 1539 certificates of Table 1, 28 and 15 cross-checks inside the new ranges by the generator and by the Python-integer checker, every certificate computed twice with identical output, a conservation-of-probability test on every run) is described in Section 8.

2 Setting

Fix integers $d \geq 1$, $N \geq 2$ and a rational $q \in (0, 1]$. Let $\Omega = \{0, \dots, N-1\}^d$ (the lattice $\{1, \dots, N\}^d$ of the article, shifted by one), $x_0 = (0, \dots, 0)$ and $a = (N-1, \dots, N-1)$ (opposite corners). For $x \in \Omega$ let $b(x)$ be the number of the $2d$ unit vectors e with $x + e \notin \Omega$. The walk $(X_t)_{t \geq 0}$ has transition matrix

$$P(x, x + e) = \frac{q}{2d} \quad (x + e \in \Omega), \quad P(x, x) = 1 - q + \frac{q}{2d} b(x),$$

that is, a move off the lattice is cancelled. P is symmetric. Let $X_0 = x_0$, $T = \min\{t \geq 1 : X_t = a\}$, $\Omega' = \Omega \setminus \{a\}$, $Q = P|_{\Omega' \times \Omega'}$, and

$$f(t) = \mathbb{P}(T = t), \quad v_t(y) = \mathbb{P}(X_{t-1} = y, T > t-1) \quad (y \in \Omega', t \geq 1), \quad \sigma_t = \sum_{y \in \Omega', |y-a|_1=1} v_t(y).$$

We write $u < w$ (resp. $u \leq w$) for vectors when the inequality holds in every coordinate.

Lemma 1 (basic identities). (a) $v_1 = e_{x_0}$ and $v_{t+1} = Qv_t$. (b) $f(t) = \frac{q}{2d} \sigma_t$. (c) $S(t) := \mathbb{P}(T > t) = \mathbf{1}^\top Q^t e_{x_0}$ and $f(t) = S(t-1) - S(t)$.

Proof. (a) $T > 0$ always and $X_0 = x_0 \neq a$, so $v_1 = e_{x_0}$. By the Markov property, for $y \in \Omega'$, $v_{t+1}(y) = \sum_{y' \in \Omega'} \mathbb{P}(X_{t-1} = y', T > t-1) P(y', y) = (Q^\top v_t)(y) = (Qv_t)(y)$, since Q is symmetric. (b) $\mathbb{P}(T = t) = \sum_{y \in \Omega'} v_t(y) P(y, a)$ and $P(y, a) = q/(2d)$ if y is a lattice neighbour of a , and 0 otherwise. (c) $S(t) = \sum_{y \in \Omega'} v_{t+1}(y) = \mathbf{1}^\top Q^t e_{x_0}$ by (a); the last identity is $\{T = t\} = \{T > t-1\} \setminus \{T > t\}$. $\square$

Lemma 2 (absorption is certain). *There are $n_0 \geq 1$ and $\delta > 0$ with $\|Q^{n_0}\|_\infty \leq 1 - \delta$. Hence the spectral radius of Q is less than 1, $\sum_{k \geq 0} Q^k = (I - Q)^{-1}$ is entrywise non-negative, $\mathbb{P}(T < \infty) = 1$ and $\mathbb{E}T = \sum_{t \geq 0} S(t) < \infty$.*

Proof. From any $y \in \Omega'$ the coordinatewise increasing lattice path to a has length at most $n_0 := d(N-1)$, avoids a before its last step, and has probability at least $\delta := (q/2d)^{n_0}$. As $Q \geq 0$, $\|Q^{n_0}\|_\infty = \max_y (Q^{n_0} \mathbf{1})(y) = \max_y \mathbb{P}_y(X_1, \dots, X_{n_0} \in \Omega') \leq 1 - \delta$. The remaining statements are the Neumann series and Lemma 1(c). $\square$

Integer form. Write $q = q_n/q_d$ in lowest terms, $g = \gcd(q_n, 2d(q_d - q_n), 2dq_d)$ and

$$c_m = \frac{q_n}{g}, \quad c_s = \frac{2d(q_d - q_n)}{g}, \quad D = \frac{2dq_d}{g}, \quad \text{so that} \quad \frac{q}{2d} = \frac{c_m}{D}, \quad 1 - q = \frac{c_s}{D}, \quad 2dc_m + c_s = D.$$

Then $M := DQ$ has non-negative integer entries: $M(y, y') = c_m$ for neighbours $y, y' \in \Omega'$, $M(y, y) = c_s + c_m b(y)$, and every row sum of M is $D - c_m \mathbf{1}[|y - a|_1 = 1] \leq D$. The nine cases used below are

d	$q = 4/5$			$q = 1/2$			$q = 1$		
	1	2	3	1	2	3	1	2	3
(c_m, c_s, D)	(2, 1, 5)	(1, 1, 5)	(2, 3, 15)	(1, 2, 4)	(1, 4, 8)	(1, 6, 12)	(1, 0, 2)	(1, 0, 4)	(1, 0, 6)

In particular $D \leq 15$ in all cases.

Definition 3. f is *strictly unimodal with mode t^** if there is $t_0 \leq t^*$ such that $f(t) = 0$ for $1 \leq t < t_0$, $f(t) < f(t+1)$ for $t_0 - 1 \leq t < t^*$ (with $f(0) := 0$), and $f(t) > f(t+1)$ for all $t \geq t^*$. Then t^* is the unique maximiser of f and f has exactly one strict local maximum (for $t < t_0 - 1$ the zero values $f(t-1) = f(t) = f(t+1) = 0$ are local maxima only in the weak sense). f is *unimodal* (in the weak sense) if there is t^* with $f(t) \leq f(t+1)$ for $t < t^*$ and $f(t) \geq f(t+1)$ for $t \geq t^*$.

3 The lemmas behind the certificates

Lemma 4 (positivity). *Every row of Q contains a positive entry. Consequently $w > 0$ implies $Qw > 0$.*

Proof. Let $y \in \Omega'$. If $q < 1$ then $Q(y, y) \geq 1 - q > 0$. If $q = 1$: among the $2d \geq 2$ directions at y at most one leads to a ; any other direction either is blocked, and then $Q(y, y) \geq q/(2d) > 0$, or leads to a site $y' \in \Omega'$, and then $Q(y, y') = q/(2d) > 0$. $\square$

Lemma 5 (tail lemma). *Suppose that for some $T \geq 1$ we have $v_{T+1} < v_T$ (at every site of Ω'). Then $v_{t+1} < v_t$ for all $t \geq T$, and*

$$f(t+1) < f(t) \quad \text{for all } t \geq T.$$

If moreover $v_{T+1} \leq \gamma v_T$ for a number $\gamma < 1$, then $f(T+s) \leq \gamma^s f(T)$ for all $s \geq 0$.

Proof. Put $w_t = v_t - v_{t+1}$. By Lemma 1(a), $w_{t+1} = Qw_t$. Since $w_T > 0$, Lemma 4 and induction give $w_t > 0$ for all $t \geq T$. By Lemma 1(b), $f(t) - f(t+1) = \frac{q}{2d} \sum_{y \sim a} w_t(y) > 0$. For the last statement, $Q \geq 0$ entrywise, so $v_{T+s+1} = Q^s v_{T+1} \leq \gamma Q^s v_T = \gamma v_{T+s}$, and the claim follows by induction and Lemma 1(b). $\square$

Lemma 5 is the device that controls the infinite tail: once all occupation probabilities are decreasing, the PMF decreases forever. It needs no spectral information. Note that a bound of the form $f(t) \leq C\rho^t$ beyond a computed horizon would *not* be enough for unimodality: it excludes a second global maximum in the tail but not a second local one. Lemma 5 gives both the monotone decrease and a geometric bound; a number $\gamma < 1$ as in its last statement always exists when the hypothesis holds, because Ω' is finite, and the verifying program records a rational upper bound for the smallest such γ .

Proposition 6 (a tail time always exists). *For all $d \geq 1$, $N \geq 2$ and real $q \in (0, 1]$ there is T with $v_{T+1} < v_T$ on Ω' . Moreover $v_{t+1}(y)/v_t(y) \rightarrow \lambda_1$ for every $y \in \Omega'$, where $\lambda_1 \in (0, 1)$ is the largest eigenvalue of Q .*

Proof. Q is symmetric and entrywise non-negative; $Q(y, y') = q/(2d) > 0$ for lattice neighbours $y, y' \in \Omega'$, and Ω' is connected by lattice steps (a path for $d = 1$, a box with one corner removed for $d \geq 2$); and $Q(x_0, x_0) = 1 - q + \frac{q}{2d}d = 1 - \frac{q}{2} > 0$, because the corner x_0 has d blocked directions. Let $\lambda_1 = \max\{u^\top Qu : \|u\|_2 = 1\}$ be the largest eigenvalue; $\lambda_1 \geq Q(x_0, x_0) > 0$. The following is the Perron–Frobenius theorem for the primitive matrix Q (Seneta, *Non-negative Matrices and Markov Chains*, 2nd ed., Theorem 1.1); for symmetric Q the proof takes a few lines, so we give it.

(1) Let u be a unit vector with $|u^\top Qu| = \lambda_1$ and let $|u|$ be its entrywise absolute value. Since $Q \geq 0$, $|u|^\top Q|u| \geq |u^\top Qu| = \lambda_1$, so $|u|$ maximises the Rayleigh quotient and $Q|u| = \lambda_1|u|$. If $|u|(y) = 0$ for some y , then $0 = \lambda_1|u|(y) = \sum_{y'} Q(y, y')|u|(y')$ forces $|u|(y') = 0$ for all neighbours y' of y , hence $|u| = 0$ by connectivity, which is impossible. So u has no zero entry.

(2) λ_1 is a simple eigenvalue: two linearly independent eigenvectors would have a non-zero linear combination vanishing at x_0 , which after normalisation contradicts (1). By (1) the eigenvector may be taken entrywise positive; call it ϕ , $\|\phi\|_2 = 1$.

(3) $-\lambda_1$ is not an eigenvalue: if $Qu = -\lambda_1 u$ with $\|u\|_2 = 1$, then $|u|^\top Q|u| = \lambda_1 = -u^\top Qu$ by (1), i.e. $\sum_{y, y'} Q(y, y')(|u(y)||u(y')| + u(y)u(y')) = 0$. Every term is non-negative and the term $y = y' = x_0$ equals $2Q(x_0, x_0)u(x_0)^2 > 0$ by (1): a contradiction.

Since $|u^\top Qu| \leq |u|^\top Q|u| \leq \lambda_1$ for unit vectors, all eigenvalues lie in $[-\lambda_1, \lambda_1]$; by (2) and (3) all eigenvalues other than λ_1 have modulus less than λ_1 . Also $\lambda_1 < 1$ by Lemma 2. Expanding e_{x_0} in an orthonormal eigenbasis of Q gives $v_t = Q^{t-1}e_{x_0} = \lambda_1^{t-1}(\phi(x_0)\phi + o(1))$ as $t \rightarrow \infty$, with $\phi(x_0)\phi > 0$ entrywise. Hence $v_{t+1}(y)/v_t(y) \rightarrow \lambda_1 < 1$ for every y , and $v_{t+1} < v_t$ for all large t . $\square$

Proposition 6 is not used in any proof below; it explains why the search for T terminates and why the recorded ratio γ is close to 1 (the modal time is long before the asymptotic regime).

Lemma 7 (enclosure by directed rounding). *Let $P \geq 1$ and $t_s \geq 1$ be integers and $X_t := M^{t-1}e_{x_0}$, an integer vector with $v_t = X_t/D^{t-1}$. Define integer vectors by*

$$L_{t_s} = \left\lfloor \frac{2^P X_{t_s}}{D^{t_s-1}} \right\rfloor, \quad U_{t_s} = \left\lceil \frac{2^P X_{t_s}}{D^{t_s-1}} \right\rceil, \quad L_{t+1} = \left\lfloor \frac{ML_t}{D} \right\rfloor, \quad U_{t+1} = \left\lceil \frac{MU_t}{D} \right\rceil \quad (t \geq t_s),$$

with floors and ceilings taken entrywise. Then for all $t \geq t_s$

$$0 \leq L_t \leq 2^P v_t \leq U_t \leq 2^P.$$

Proof. For $t = t_s$ this is the definition and $0 \leq v_t \leq 1$. If it holds for t , then, because $M \geq 0$ entrywise, $ML_t/D \leq 2^P Qv_t = 2^P v_{t+1} \leq MU_t/D$, and floor and ceiling preserve the inequalities because they are monotone and fix the bound on the appropriate side. Finally every row sum of M is at most D , so each entry of MU_t/D is at most $\max_y U_t(y) \leq 2^P$, an integer; hence so is its ceiling. $\square$

Lemma 8 (exactness of the limb engine). *Let $D \leq 15$, $\beta = 2^{58}$, and let a non-negative integer vector $Z \leq 2^P$ with $P = 112$ be stored as two `int64` arrays $(Z^{(0)}, Z^{(1)})$, $Z = Z^{(0)} + \beta Z^{(1)}$, $0 \leq Z^{(0)} < \beta$, $0 \leq Z^{(1)} \leq 2^{54}$. The procedure `step` of `fp limb.py` computes $\lfloor MZ/D \rfloor$ (and, when $D - 1$ is first added to the low limb, $\lceil MZ/D \rceil$) exactly, and no intermediate `int64` value leaves $[0, 2^{63})$.*

Proof. The stencil weights are non-negative and sum to at most D , so the limbwise images satisfy $0 \leq W^{(0)} \leq D(\beta - 1)$ and $0 \leq W^{(1)} \leq D2^{54}$, and $MZ = W^{(0)} + \beta W^{(1)}$. With $W^{(1)} = Dq_1 + r_1$, $0 \leq r_1 < D$, and $c = r_1\beta + W^{(0)}(+D - 1) = Dq_0 + r_0$ we get $MZ(+D - 1) = D(q_0 + \beta q_1) + r_0$ with $0 \leq r_0 < D$, i.e. $q_0 + \beta q_1$ is the required floor (ceiling). The largest intermediate value is $c \leq (D - 1)\beta + D(\beta - 1) + D - 1 < 2D\beta \leq 30 \cdot 2^{58} < 2^{63}$. The quotient digit $q_0 < 2\beta$ is normalised by one carry into q_1 ; by Lemma 7 the result is at most 2^P , so $q_1 \leq 2^{54}$ after the carry. $\square$

Lemma 9 (one vector and a scalar error bound). *Let $P \geq 1$, let K be an integer with $2^K > D 2^P$, $R = \lfloor 2^K/D \rfloor$ and $t_s \geq 1$. Define integer vectors C_t and integers b_t by*

$$C_{t_s} = \left\lfloor \frac{2^P X_{t_s}}{D^{t_s-1}} \right\rfloor, \quad b_{t_s} = 1, \quad C_{t+1} = \left\lfloor \frac{(MC_t)R}{2^K} \right\rfloor, \quad b_{t+1} = b_t + 2 \quad (t \geq t_s).$$

Then for all $t \geq t_s$, entrywise on Ω' ,

$$0 \leq C_t \leq 2^P v_t \leq C_t + b_t, \quad C_t \leq 2^P.$$

Proof. For $t = t_s$ this is the definition of the floor and $0 \leq v_t \leq 1$. Assume the claim for t and put $W = MC_t$, a non-negative integer vector with entries at most $D 2^P$ (the row sums of M are at most D). Since $2^K/D - 1 < R \leq 2^K/D$,

$$\frac{W}{D} \geq \frac{WR}{2^K} \geq \frac{W}{D} - \frac{W}{2^K} \geq \frac{W}{D} - \frac{D 2^P}{2^K} > \frac{W}{D} - 1,$$

so that $W/D - 2 < C_{t+1} \leq W/D$. Because $Q = M/D$ is entrywise non-negative with row sums at most 1,

$$C_{t+1} \leq QC_t \leq Q(2^P v_t) = 2^P v_{t+1} \leq QC_t + b_t Q \mathbf{1} \leq \frac{W}{D} + b_t < C_{t+1} + 2 + b_t,$$

and $C_{t+1} \leq 2^P v_{t+1} \leq 2^P$. □

Lemma 9 needs half the work of Lemma 7 (one vector instead of two) and uses a multiplication by a reciprocal instead of a division; the price is that the upper bound $C_t + b_t$ grows by 2 units of 2^{-P} per step at every site, whether needed or not. With $P = 112$ and fewer than 10^6 steps this is harmless.

Lemma 10 (packed integer arithmetic). *Let $w \geq 1$, $n = N^d$, and number the sites by $i(y) = \sum_{k=1}^d y_k N^{k-1} \in \{0, \dots, n-1\}$, so that $i(x_0) = 0$ and $i(a) = n-1$. For a vector z on Ω with integer entries $0 \leq z(y) < 2^w$ let $\langle z \rangle = \sum_y z(y) 2^{wi(y)}$ (a single non-negative integer; the w bits numbered $w i(y), \dots, w i(y) + w - 1$ are the field of y). Let $\&$ denote the bitwise and of non-negative integers, and for $k = 1, \dots, d$ put $s_k = w N^{k-1}$,*

$$H_k = \langle (2^w - 1) \mathbf{1}_{[y_k \leq N-2]} \rangle, \quad W_k = \langle (2^w - 1) \mathbf{1}_{[y_k \in \{0, N-1\}]} \rangle, \quad E = \sum_{y \neq a} 2^{wi(y)}.$$

(a) (stencil) *If $z(a) = 0$ and $D \max_y z(y) < 2^w$, then, with Mz extended by 0 at a ,*

$$\langle Mz \rangle = \left(c_m \sum_{k=1}^d \left[(\langle z \rangle \& H_k) 2^{s_k} + (\lfloor \langle z \rangle / 2^{s_k} \rfloor \& H_k) + (\langle z \rangle \& W_k) \right] + c_s \langle z \rangle \right) \& (2^{w(n-1)} - 1).$$

(b) (comparison at every site) *Let u, z be integer vectors with $u(a) = z(a) = 0$ and $0 \leq u, z \leq \beta$ entrywise, and let $c \geq 0$ be an integer with $\beta + c + 1 < 2^{w-1}$. Put $V = \langle z \rangle + 2^{w-1} E - \langle u \rangle - (c+1)E$. Then*

$$u(y) + c < z(y) \quad \text{for all } y \in \Omega' \quad \Longleftrightarrow \quad V \& 2^{w-1} E = 2^{w-1} E.$$

(c) (division by reciprocal multiplication) *If $R \geq 1$ and $0 \leq K < w$ are integers and $z(y)R < 2^w$ for all y , then*

$$\langle \lfloor zR/2^K \rfloor \rangle = \lfloor \langle z \rangle R/2^K \rfloor \& \langle (2^{w-K} - 1) \mathbf{1} \rangle.$$

Proof. Two facts are used throughout: if $0 \leq z_1(y) + z_2(y) < 2^w$ for all y , then $\langle z_1 \rangle + \langle z_2 \rangle = \langle z_1 + z_2 \rangle$ (no carry leaves a field); and $\langle z \rangle \& \langle (2^w - 1)\mathbf{1}_S \rangle = \langle z\mathbf{1}_S \rangle$ for every set S of sites.

(a) $\langle z \rangle \& H_k = \langle z\mathbf{1}[y_k \leq N - 2] \rangle$, and multiplying by 2^{s_k} moves the field of y to position $i(y) + N^{k-1} = i(y + e_k)$, which is a site of Ω because $y_k \leq N - 2$; so the first term is $\langle u_1 \rangle$ with $u_1(y) = z(y - e_k)\mathbf{1}[y_k \geq 1]$. Next, $\sum_{i < m} z_i 2^{wi} < 2^{wm}$ for every m , so $\lfloor \langle z \rangle / 2^{s_k} \rfloor = \sum_{i \geq N^{k-1}} z_i 2^{w(i - N^{k-1})}$: the field at position j now holds $z_{j + N^{k-1}}$. The and with H_k keeps the positions $j = i(y)$ with $y_k \leq N - 2$, for which $j + N^{k-1} = i(y + e_k)$; the positions with $y_k = N - 1$, which received the entry of the site $(\dots, 0, y_{k+1} + 1, \dots)$ of the next row, are cleared. So the second term is $\langle u_2 \rangle$ with $u_2(y) = z(y + e_k)\mathbf{1}[y_k \leq N - 2]$. The third term is $\langle z(\mathbf{1}[y_k = 0] + \mathbf{1}[y_k = N - 1]) \rangle$ (two different sets, as $N \geq 2$): one cancelled move in direction $\mp e_k$. Summing over k gives, at each $y \in \Omega$, $\sum_{y' \sim y} z(y') + b(y)z(y)$; multiplying by c_m and adding $c_s z(y)$ gives a number at most $(2dc_m + c_s) \max z = D \max z < 2^w$, so no carry occurs and the big parenthesis is the packing of this vector. For $y \in \Omega'$ its entry is $(Mz)(y)$, because $z(a) = 0$. The field of a (the mass absorbed in this step) is the highest field and is cleared by the last and.

(b) For $y \in \Omega'$ put $\alpha_y = z(y) - u(y) - c - 1$; then $-2^{w-1} < -\beta - c - 1 \leq \alpha_y \leq \beta < 2^{w-1}$, so $V = \sum_{y \in \Omega'} (2^{w-1} + \alpha_y) 2^{wi(y)}$ with every coefficient in $[0, 2^w)$: the field of y in V is exactly $2^{w-1} + \alpha_y$, and its top bit is set if and only if $\alpha_y \geq 0$, i.e. $u(y) + c < z(y)$.

(c) $\langle z \rangle R = \langle zR \rangle$ by hypothesis. Writing $z(y)R = 2^K \lfloor z(y)R/2^K \rfloor + \rho_y$ with $0 \leq \rho_y < 2^K$, $\lfloor \langle zR \rangle / 2^K \rfloor = \sum_y \lfloor z(y)R/2^K \rfloor 2^{wi(y)} + \sum_{i(y) \geq 1} \rho_y 2^{w i(y) - K}$. The first sum occupies the low $w - K$ bits of each field (as $\lfloor z(y)R/2^K \rfloor < 2^{w-K}$), the second the top K bits of the field below; the and removes the second. $\square$

Certificate data. Two programs compute, by different routes, the following data for given (d, N, q) and $P = 112$.

- The exact integers $s_t = \sum_{y \sim a} X_t(y)$ for $1 \leq t \leq t_s$, so that $\sigma_t = s_t / D^{t-1}$. Here t_s is the first t with $\lfloor 2^P s_t / D^{t-1} \rfloor \geq 2^G$ for a guard parameter G (any t_s gives valid enclosures; the choice only ensures that the rounded phase starts with G significant bits).
- For every t an integer enclosure $F_t^- \leq 2^P \sigma_t \leq F_t^+$: for $t < t_s$ the floor and the ceiling of $2^P s_t / D^{t-1}$; for $t \geq t_s$ the sums over the d neighbours of a of the entrywise bounds of Lemma 7 (generator: $F_t^- = \sum_{y \sim a} L_t(y)$, $F_t^+ = \sum_{y \sim a} U_t(y)$, $G = 48$) or of Lemma 9 (checker: $F_t^- = \sum_{y \sim a} C_t(y)$, $F_t^+ = F_t^- + db_t$, $G = 44$, $K = 120$).
- A certified relation $\text{rel}_t \in \{<, >, =, ?\}$ between $f(t)$ and $f(t + 1)$: for $t < t_s$ by the exact comparison of Ds_t with s_{t+1} ; for $t \geq t_s$, “ $<$ ” if $F_t^+ < F_{t+1}^-$, “ $>$ ” if $F_t^- > F_{t+1}^+$, and “ $?$ ” otherwise.
- The *tail time* T : the first t at which $v_{t+1} < v_t$ is certified at every site of Ω' , i.e. $U_{t+1} < L_t$ (generator), $C_{t+1} + b_{t+1} < C_t$ (checker), or the exact condition $X_{t+1} < DX_t$ if this happens before t_s .
- (Checker only.) A rational $\gamma < 1$ with $v_{T+1} \leq \gamma v_T$: the maximum over $y \in \Omega'$ of $(C_{T+1}(y) + b_{T+1}) / C_T(y)$, resp. of $X_{T+1}(y) / (DX_T(y))$.

Proposition 11 (what a certificate proves). *Suppose that the tail time T exists, that $F_t^- \leq 2^P \sigma_t \leq F_t^+$ for $1 \leq t \leq T + 1$, and that each rel_t , $1 \leq t < T$, is either “ $?$ ” or a true relation between $f(t)$ and $f(t + 1)$. Let $t^* \in \{1, \dots, T\}$ maximise F_t^- .*

(i) $f(t + 1) < f(t)$ for all $t \geq T$, and $f(T + s) \leq \gamma^s f(T)$ for all $s \geq 0$ if $v_{T+1} \leq \gamma v_T$.

(ii) If $F_{t^*}^- > F_t^+$ for all $t \in \{1, \dots, T\} \setminus \{t^*\}$, then $f(t^*) > f(t)$ for every $t \neq t^*$: the mode is unique and equals t^* .

- (iii) Let t_0 be the first t with $\sigma_t > 0$. If, in addition to (ii), rel_t is “<” for $\max(t_0 - 1, 1) \leq t < t^*$ and “>” for $t^* \leq t < T$, then f is strictly unimodal with mode t^* in the sense of Definition 3. (The two programs of this section, which have an exact phase, find t_0 exactly: $t_0 \leq t_s$, and for $t \leq t_s$ the bound F_t^+ is the ceiling of the exact value, so $F_t^+ > 0 \iff \sigma_t > 0$. For the C engine of Section 8, which has no exact phase, the same conclusion $F_t^+ > 0 \iff \sigma_t > 0$ is part of Proposition 38.)
- (iv) Call $t \in \{1, \dots, T\}$ a certified peak if ($t = 1$ or rel_{t-1} is “<”) and ($t = T$ or rel_t is “>”). If there are $k \geq 2$ certified peaks, then f has at least k strict local maxima; in particular it is not unimodal.

Proof. (i) is Lemma 5. (ii) For $t \leq T$ the hypothesis gives $2^P \sigma_{t^*} \geq F_{t^*}^- > F_t^+ \geq 2^P \sigma_t$, and $f = \frac{q}{2d} \sigma$. For $t > T$, (i) gives $f(t) < f(T) \leq f(t^*)$. (iii) f vanishes before t_0 by definition, and $t_0 \leq t^*$ because $f(t^*) = \max f > 0$ by (ii). The relations give $f(t) < f(t+1)$ for $t_0 - 1 \leq t < t^*$ (for $t_0 = 1$ the first of these is $f(0) = 0 < f(1)$) and $f(t) > f(t+1)$ for $t^* \leq t < T$; (i) gives the rest of Definition 3. (iv) Each certified peak satisfies $f(t-1) < f(t) > f(t+1)$ (with $f(0) = 0$; for $t = 1$ note that $f(1) > f(2) \geq 0$, and for $t = T$ use (i)). $\square$

The hypotheses of Proposition 11 hold for the data of both programs: the enclosures by Lemma 7, resp. Lemma 9; the relations because $\sigma_t < \sigma_{t+1} \iff Ds_t < s_{t+1}$ in the exact phase and by the enclosures afterwards; the tail time because $2^P v_{T+1} \leq U_{T+1} < L_T \leq 2^P v_T$, resp. $2^P v_{T+1} \leq C_{T+1} + b_{T+1} < C_T \leq 2^P v_T$ (in the exact phase $X_{T+1} < DX_T$ is literally $v_{T+1} < v_T$).

The programs.

- *Generator* `fplimb.py`: the recurrences of Lemma 7 in the limb arithmetic of Lemma 8 (its exact phase is the Python-integer code of `fpcore.py`). It writes one JSON line per (d, N, q) with t_s, t_0, t^*, T , the verdicts of (ii)–(iv), the enclosures $F_t^\pm$ for $t = t^* - 1, t^*, t^* + 1$, the smallest margin $\min_y (L_T - U_{T+1})(y)$ and the sha256 digest of the two integer vectors L_T, U_{T+1} . These lines are the certificates `certs/modes_*.jsonl`.
- *Checker* `verify_packed.py` (about 250 lines including comments, standard library only): the recurrence of Lemma 9 with the whole vector packed into one Python integer as in Lemma 10, with $w = 240$ in the rounded phase ($2^w > 2^{P+K} \geq z(y)R$, as required by (c)) and a field width that grows with t in the exact phase. The tail condition is tested with Lemma 10(b) and, when the test succeeds, re-checked site by site after unpacking. It re-derives all verdicts from scratch, writes them to `certs/packed_*.jsonl` and compares them with the generator’s line: t_0, t^*, T , the verdicts of (ii) and (iii) and the number of certified peaks must be identical, and the enclosures of $2^P \sigma_t$ at $t^* - 1, t^*, t^* + 1$ (same scale 2^P) must intersect. With the option `--exact` it never leaves the exact phase: then $F_t^- = F_t^+ = 2^P \sigma_t$ are exact rationals, no relation is “?”, and all verdicts, including ties, are exact (`certs/packedexact_*.jsonl`).
- *Reference* `fpcore.py`: the recurrences of Lemma 7 with Python integers; where it was run, its output is identical to the generator’s, digest included (`certs/refcheck_*.jsonl`).
- `verify_exact.py` (about 100 lines): builds the chain site by site from the rules of Section 2 and iterates it with exact rationals; feasible for small N only (`certs/exactcheck_*.jsonl`).

d	q	certified N (generator <code>fplimb</code>)	number	verified Python (with <code>verify_packed</code>) integers	exact arithmetic	bit-identical (<code>fpcore</code>)
1	4/5	2–600, 800, 1000	601	all	2–160	2–200, 250, 300, 350, 400, 450, 500, 550, 600
1	1/2	2–300	299	all	2–120	2–120
1	1	2–200	199	all	2–100	2–100
2	4/5	2–160, 180, 200	161	all	2–50	2–60, 70, 80, 90, 100
2	1/2	2–80	79	all	2–40	2–40
2	1	2–80	79	all	2–40	2–40
3	4/5	2–60	59	all	2–20	2–24, 28, 32, 36
3	1/2	2–32	31	all	2–16	2–16
3	1	2–32	31	all	2–16	2–16

Table 1: Ranges of N covered by the mode certificates, and the parts re-derived by each of the separately written programs (all written within this project and run on the same machine). “Verified with Python integers”: `verify_packed.py` reproduced t_0 , t^* , T and all verdicts, and its enclosures intersect the generator’s. “Exact arithmetic”: `verify_packed.py --exact` or `verify_exact.py`; no rounding occurs, ties are decided. “Bit-identical”: `fpcore.py` reproduced the generator’s JSON line including the digest of (L_T, U_{T+1}) .

4 Certified modes and unimodality

For $d \in \{1, 2, 3\}$ and $q \in \{4/5, 1/2, 1\}$ let $\mathcal{N}(d, q)$ be the set of N in the third column of Table 1:

	$d = 1$	$d = 2$	$d = 3$
$q = 4/5$	$2 \leq N \leq 600$ and $N \in \{800, 1000\}$	$2 \leq N \leq 160$ and $N \in \{180, 200\}$	$2 \leq N \leq 60$
$q = 1/2$	$2 \leq N \leq 300$	$2 \leq N \leq 80$	$2 \leq N \leq 32$
$q = 1$	$2 \leq N \leq 200$	$2 \leq N \leq 80$	$2 \leq N \leq 32$

Theorem 12 (modes and unimodality for $q = 4/5$ and $q = 1/2$; computer-assisted). *Let $d \in \{1, 2, 3\}$, $q \in \{4/5, 1/2\}$ and $N \in \mathcal{N}(d, q)$. Call (d, q, N) exceptional if it is one of $(1, \frac{4}{5}, 4)$, $(1, \frac{1}{2}, 3)$, $(1, \frac{1}{2}, 4)$.*

- (a) $f(t) = 0$ for $t < d(N - 1)$ and $f(d(N - 1)) > 0$.
- (b) If (d, q, N) is not exceptional, f is strictly unimodal in the sense of Definition 3: it has exactly one strict local maximum, at the integer $t^* = t^*(d, N, q)$ given in the field *mode* of line N of `certs/modes_dd_q4-5.jsonl` (resp. `..._q1-2.jsonl`), with $f(t) < f(t + 1)$ for $d(N - 1) - 1 \leq t < t^*$ and $f(t) > f(t + 1)$ for all $t \geq t^*$. Tables 2–4 list a selection of the modes, Appendix A lists all of them for $d = 2, 3$, and Theorem 30 gives a formula for $d = 1$.
- (c) If (d, q, N) is not exceptional, every occupation probability is already decreasing at the modal time: $v_{t^*+1} < v_{t^*}$ on Ω' (the tail time of Proposition 11 is $T = t^*$). Moreover $f(t^* + s) \leq \gamma^s f(t^*)$ for all $s \geq 0$, where $\gamma = \gamma(d, N, q) < 1$ satisfies $1 - \gamma \geq$ the number in the field `one_minus_gamma_lower` of `certs/packed_*.jsonl`.
- (d) The exceptional cases are unimodal in the weak sense, with an exact tie: $(1, \frac{4}{5}, 4)$: $0 = f(2) < f(3) = f(4) < f(5) > f(6) > \dots$, unique mode $t^* = 5$, $T = 5$; $(1, \frac{1}{2}, 3)$: $0 = f(1) < f(2) < f(3) = f(4) > f(5) > \dots$, maximum attained exactly at $t \in \{3, 4\}$, $T = 4$; $(1, \frac{1}{2}, 4)$: f strictly increasing on $2 \leq t \leq 7$, $f(7) = f(8)$, strictly decreasing from 8 on, maximum attained exactly at $t \in \{7, 8\}$, $T = 8$.

Proof. For each (d, q, N) the certificate line was produced by `fplimb.certify`. By Lemma 8 the integers it manipulates are those of Lemma 7, and the recorded verdicts are the hypotheses of Proposition 11(i)–(iii); $t_0 = d(N - 1)$ and $T = t^*$ are read off the certificates (the value of t_0 is also

given, for all d, N, q , by Lemma 34). For every N (fifth column of Table 1) the same verdicts were obtained by the separately written program `verify_packed.py`, which rests on Lemmas 9 and 10 and on Python integers only, and which also supplies γ in (c); so the theorem does not depend on Lemma 8 or on numpy. The exceptional cases, in which the rounded computations necessarily return “?” for one relation, are settled in exact arithmetic by `verify_packed.py --exact` and, with separately written code, by `verify_exact.py` (sixth column); the three displayed chains can also be checked by hand from $f(t) = \frac{q}{2}(Q^{t-1})_{N-2,0}$. The script `scripts/audit.py` re-checks every assertion of the theorem against the certificate files. $\square$

Theorem 13 (modes and multimodality for $q = 1$; computer-assisted). *Let $q = 1$, $d \in \{1, 2, 3\}$ and $N \in \mathcal{N}(d, 1)$. Then $f(t) = 0$ exactly for $t < d(N - 1)$, the smallest maximiser of f is the integer in the field `mode` of `certs/modes_dd_q1-1.jsonl`, and:*

- $d = 1$: *the maximiser is unique except for $N \in \{5, 6, 8, 9\}$, where the maximum is attained at exactly two times ($N = 5$: $t \in \{4, 6\}$; $N = 6$: $t \in \{7, 9\}$; $N = 8$: $t \in \{15, 17\}$; $N = 9$: $t \in \{20, 22\}$); f is strictly unimodal for $N \in \{2\}$; f has at least two strict local maxima, hence is not unimodal, for $N \in \{4-200\}$ (the largest certified number of strict local maxima is 37127); for $N = 3$, f is unimodal in the weak sense but not strictly: $f(3) = f(4)$; equal consecutive non-zero values occur only for $N = 3$ ($f(3) = f(4)$) and $N = 4$ ($f(8) = f(9)$) (exact arithmetic): for every other pair (N, t) with N in the range and $t \geq d(N - 1) - 1$, $f(t) \neq f(t + 1)$; the tail time T exceeds t^* for 198 of the 199 values of N (largest ratio $T/t^* = 5.64$).*
- $d = 2$: *the maximiser is unique except for $N \in \{2\}$, where the maximum is attained at exactly two times ($N = 2$: $t \in \{2, 3\}$); f is strictly unimodal for $N \in \{3-8, 10, 12, 14, 16, 19-20, 22-23, 25, 27-37, 39-50, 52-60, 62-80\}$; f has at least two strict local maxima, hence is not unimodal, for $N \in \{9, 11, 13, 15, 17-18, 21, 24, 26, 38, 51, 61\}$ (the largest certified number of strict local maxima is 2); for $N = 2$, f is unimodal in the weak sense but not strictly: $f(2) = f(3)$; equal consecutive non-zero values occur only for $N = 2$ ($f(2) = f(3)$) (exact arithmetic): for every other pair (N, t) with N in the range and $t \geq d(N - 1) - 1$, $f(t) \neq f(t + 1)$; the tail time T exceeds t^* for 40 of the 79 values of N (largest ratio $T/t^* = 1.50$).*
- $d = 3$: *the maximiser is unique for every N of the range; f is strictly unimodal for $N \in \{2-32\}$; $f(t) \neq f(t + 1)$ for every N of the range and every $t \geq d(N - 1) - 1$.*

Proof. As for Theorem 12, now using also Proposition 11(iv). The statements about ties are exact computations (`verify_packed.py --exact` and `verify_exact.py`). $\square$

Remark 14 (margins). The certified separation is far from marginal. In every certificate with a unique maximiser the gap between the lower bound for $2^P \sigma_{t^*}$ and the upper bounds at $t^* \pm 1$ exceeds the width of the generator’s enclosure by a factor of at least 10^{11} . With $P = 112$ the generator’s enclosure width $U_t - L_t$ at the mode does not exceed 27219 units of 2^{-112} and the checker’s error bound b_T does not exceed 801427 units in any run, while $2^P \sigma_{t^*}$ is of order $10^{27}-10^{33}$.

Remark 15 (the geometric ratio). The ratios γ of part (c) are close to 1: the smallest recorded value of $1 - \gamma$ over the certified N is 1.41×10^{-12} , 2.22×10^{-14} , 1.76×10^{-12} for $d = 1, 2, 3$ at $q = 4/5$ (and 1.25×10^{-11} , 5.91×10^{-11} , 1.09×10^{-10} at $q = 1/2$). This is not a defect of the method: $T = t^*$, and at the modal time the neighbours of the target have only just begun to lose probability ($f(t^* + 1)/f(t^*)$ is itself within about t^{*-2} of 1), so the best constant in $v_{T+1} \leq \gamma v_T$ is barely below 1. The bound suffices for what it is used for (the tail is decreasing and summable with an explicit rate); by Proposition 6 the true decay rate of f is λ_1 , and a sharper γ could be certified by applying Lemma 5 at a later time.

Remark 16 (agreement with the floating-point study). The certified integers coincide with the values obtained in double precision in the computations of Sections 5–6 of the article (`data/msc_modes/discrete_modes.jsonl`) for all 88 corner-to-corner cases at $q \in \{4/5, 1/2, 1\}$

N	$t^* (q = 4/5)$	$\tau_N = q \mathbb{E}T$	$t^*/\mathbb{E}T$	t_{cf}	$(t_{\text{cf}} - t^*)/t^*$	$t^* (q = 1/2)$	$t^* (q = 1)$
2	1	2.0000	0.400000	1.155	+0.15525	1	1
3	2	6.0000	0.266667	3.360	+0.67977	3	2
5	8	20.0000	0.320000	10.547	+0.31837	13	4
10	37	90.0000	0.328889	45.289	+0.22404	59	27
20	158	380.0000	0.332632	187.050	+0.18386	253	119
35	495	1190.0000	0.332773	580.503	+0.17273	793	384
50	1020	2450.0000	0.333061	1190.946	+0.16759	1633	799
100	4124	9900.0000	0.333253	4792.935	+0.16221	6599	3267
200	16580	39800.0000	0.333266	19230.068	+0.15984	26529	13199
300	37369	89700.0000	0.333280	43311.413	+0.15902	59791	—
400	66490	159600.0000	0.333283	77036.970	+0.15862	—	—
500	103943	249500.0000	0.333284	120406.740	+0.15839	—	—
600	149727	359400.0000	0.333282	173420.722	+0.15825	—	—
800	266294	639200.0000	0.333284	308381.324	+0.15805	—	—
1000	416188	999000.0000	0.333284	481918.777	+0.15794	—	—

Table 2: $d = 1$: selected certified modes. Columns 3–6 refer to $q = 4/5$; here $\tau_N = N(N - 1)$ exactly, and $t^*/\mathbb{E}T$, t_{cf} and the relative error are rounded values of certified enclosures (Section 5).

N	$t^* (q = 4/5)$	$\tau_N = q \mathbb{E}T$	$t^*/\mathbb{E}T$	t_{cf}	$(t_{\text{cf}} - t^*)/t^*$	$t^* (q = 1/2)$	$t^* (q = 1)$
2	3	8.0000	0.300000	3.961	+0.32028	5	2
3	10	27.0000	0.296296	11.409	+0.14087	17	8
5	36	106.8182	0.269617	37.514	+0.04206	58	28
10	172	602.3339	0.228445	173.028	+0.00598	275	138
20	765	3113.8124	0.196544	761.935	−0.00401	1225	612
35	2493	11280.6464	0.176798	2473.311	−0.00790	3989	1994
50	5253	25291.8381	0.166156	5206.194	−0.00891	8405	4202
80	13939	72406.1447	0.154009	13806.397	−0.00951	22303	11150
100	22110	118816.6057	0.148868	21897.610	−0.00961	—	—
120	32209	177781.2819	0.144938	31897.036	−0.00969	—	—
140	44248	249673.6990	0.141779	43818.943	−0.00970	—	—
160	58239	334809.1538	0.139157	57675.051	−0.00968	—	—
180	74192	433460.4919	0.136930	73475.274	−0.00966	—	—
200	92116	545868.2190	0.135001	91228.178	−0.00964	—	—

Table 3: $d = 2$: selected certified modes (same conventions; τ_N rounded).

N	$t^* (q = 4/5)$	$\tau_N = q \mathbb{E}T$	$t^*/\mathbb{E}T$	t_{cf}	$(t_{\text{cf}} - t^*)/t^*$	$t^* (q = 1/2)$	$t^* (q = 1)$
2	8	20.0000	0.320000	7.905	−0.01191	12	6
3	24	82.9286	0.231525	24.332	+0.01385	39	19
5	85	444.2662	0.153061	85.663	+0.00780	137	68
10	422	3933.1764	0.085834	423.160	+0.00275	676	337
15	1044	13708.3719	0.060926	1045.164	+0.00112	1671	835
20	1967	33010.2340	0.047670	1968.259	+0.00064	3148	1573
30	4759	113155.3525	0.033646	4760.577	+0.00033	7615	3807
35	6644	180479.3005	0.029450	6645.582	+0.00024	—	—
40	8863	270291.3016	0.026232	8864.794	+0.00020	—	—
50	14323	530340.8436	0.021606	14325.785	+0.00019	—	—
60	21175	919226.7397	0.018429	21178.048	+0.00014	—	—

Table 4: $d = 3$: selected certified modes (same conventions).

that fall in the certified sets (0 disagreements); for example $d = 1$: $N = 100, 200, 600, 1000$ give 4124, 16580, 149727, 416188; $d = 2$: $N = 35, 100, 160, 200$ give 2493, 22110, 58239, 92116; $d = 3$: $N = 35, 40, 60$ give 6644, 8863, 21175 (all at $q = 4/5$). The remark of Sections 5–6 of the article that no exact tie occurred in their runs is consistent with part (d): the ties occur only for $N \leq 4$ in one dimension (and for $q = 1$), which were not among its cases.

Remark 17 (the exceptional cases). $(1, \frac{4}{5}, 4)$ is the boundary case of a general phenomenon: for $d = 1$ the first two non-zero values are $f(N-1) = (q/2)^{N-1}$ and $f(N) = (q/2)^{N-1}[(1 - \frac{q}{2}) + (N-2)(1-q)]$ (one resting step inserted at one of the $N-1$ sites, the reflecting end having resting probability $1 - \frac{q}{2}$), so that $f(N) \geq f(N-1)$ if and only if $q \leq (2N-4)/(2N-3)$, with equality for $N = 4$ exactly at $q = 4/5$. The two ties at $q = 1/2$ are plateaus at the top, which Theorem 28 allows.

Corollary 18 (the mode scales like $1/q$ up to one time step; computer-assisted). *Let $t^*(q)$ denote the smallest maximiser of f for the activity q , and $\Delta_N := \frac{4}{5}t^*(\frac{4}{5}) - \frac{1}{2}t^*(\frac{1}{2})$. Then $-0.7 \leq \Delta_N \leq 0.5$ for $d = 1$, $2 \leq N \leq 300$; $-0.8 \leq \Delta_N \leq 0.3$ for $d = 2$, $2 \leq N \leq 80$; $-1.1 \leq \Delta_N \leq 0.4$ for $d = 3$, $2 \leq N \leq 32$; and all six bounds are attained.*

Proof. Exact arithmetic on the integers of Theorem 12 (`scripts/summarise.py`, re-checked by `scripts/audit.py`). $\square$

So in the certified ranges $qt^*(q)$ is the same for $q = 4/5$ and $q = 1/2$ up to about one unit, although the modes themselves reach 59791 ($d = 1$, $N = 300$, $q = 1/2$): the discrete mode is, to this accuracy, a function of qt only. This fails at $q = 1$ in one dimension, where the parity oscillation moves the raw $\arg\max(\frac{4}{5}t^*(\frac{4}{5}) - t^*(1))$ reaches 66 for $N \leq 200$, at $N = 199$.

5 The mean first-passage time and the closed form

Lemma 19 (rational enclosure of the MFPT). *Let $m(x) = \mathbb{E}_x[T_a]$ for $x \in \Omega'$, where T_a is the hitting time of a . Then $\tau := qm$ does not depend on q and is the unique solution of*

$$K\tau = 2d\mathbf{1}, \quad K := 2dI - (A + B) \quad \text{on } \Omega',$$

where A is the adjacency matrix of the box restricted to Ω' and $B = \text{diag}(b)$. Moreover $K^{-1} \geq 0$ entrywise. Consequently, for any real vector z on Ω' such that $R := Kz > 0$ entrywise,

$$\frac{2d}{\max_y R(y)} z \leq \tau \leq \frac{2d}{\min_y R(y)} z \quad \text{entrywise.}$$

Proof. Let Q_1 be the matrix Q for $q = 1$, so that $Q = (1-q)I + qQ_1$ and $2d(I - Q_1) = K$. By Lemma 2 (applied with $q = 1$), $(I - Q_1)^{-1} = \sum_k Q_1^k$ is entrywise non-negative, and so is $K^{-1} = (2d)^{-1}(I - Q_1)^{-1}$. By Lemma 2 for the given q (its proof applies to every starting site), $m = \sum_{t \geq 0} Q^t \mathbf{1} = (I - Q)^{-1} \mathbf{1}$, because $\mathbb{E}_x T_a = \sum_{t \geq 0} \mathbb{P}_x(T_a > t)$; that is, $q(I - Q_1)m = \mathbf{1}$, i.e. $K\tau = 2d\mathbf{1}$. Finally $K(\frac{2d}{\min R} z - \tau) = 2d(\frac{R}{\min R} - \mathbf{1}) \geq 0$, and applying $K^{-1} \geq 0$ gives the upper bound; the lower bound is symmetric. $\square$

Use. `scripts/mfpt_enclosure.py` obtains a guess z in floating point (cosine-transform solve of the reflecting problem plus three refinement steps; untrusted), rounds it to an integer vector at scale 2^{96} , computes $R = Kz$ exactly in Python integers, and outputs the rational interval $[2dz(x_0)/\max R, 2dz(x_0)/\min R] \ni \tau_N := \tau(x_0) = q\mathbb{E}_{x_0} T$. The relative width is below 10^{-28} in every case (and zero in the cases where the rounded guess is exact). Checks: for $d = 1$ the enclosures contain $N(N-1)$ (the exact value, since $m(j) = [N(N-1) - j(j+1)]/q$ solves the system); for $d = 2$ they contain the exact rationals 8, 27, 416/7, 1175/11, 9368/55 ($N = 2, \dots, 6$), and they overlap the Arb evaluation of the single-sum formula of the companion note for all $N \leq 60$, every twentieth N up to 300 and the five isolated values $N = 350, 401, 450, 500, 600$.

Definition 20. For d, N, q as above put $\mu_1 = \frac{q}{d}(1 - \cos \frac{\pi}{N})$, $X_N = \mu_1 \mathbb{E}_{x_0} T = \frac{\tau_N}{d}(1 - \cos \frac{\pi}{N})$ (independent of q) and

$$t_{\text{cf}} = t_{\text{cf}}(d, N, q) = \frac{\tau_N}{q} \cdot \frac{\ln(2dX_N)}{X_N + 2d - 1}.$$

Theorem 21 (accuracy of the closed form; computer-assisted). *Let t^* be the certified mode of Theorem 12 and t_{cf} as in Definition 20.*

- (a) $d = 2$: for $q = 4/5$ and all $10 \leq N \leq 160$, $|t^* - t_{\text{cf}}| \leq 0.00970 t^*$; for $q = 1/2$ and all $10 \leq N \leq 80$, $|t^* - t_{\text{cf}}| \leq 0.00954 t^*$.
- (b) $d = 3$: for $q = 4/5$ and all $10 \leq N \leq 60$, $|t^* - t_{\text{cf}}| \leq 0.00275 t^*$; for $q = 1/2$ and all $10 \leq N \leq 32$, $|t^* - t_{\text{cf}}| \leq 0.00157 t^*$.
- (c) $d = 1$: for $q = 4/5$ and all $10 \leq N \leq 600$, $0.1582 t^* \leq t_{\text{cf}} - t^* \leq 0.2251 t^*$; for $q = 1/2$ and all $10 \leq N \leq 300$, $0.1590 t^* \leq t_{\text{cf}} - t^* \leq 0.2282 t^*$. The closed form therefore overestimates the one-dimensional mode by more than 15% throughout the range; it is not an approximation of the mode in one dimension.

Sharper bounds on sub-ranges, the bounds for the isolated larger values of N , and the corresponding numbers for $q = 1$ are given in Table 5; the enclosure of the relative error for every single N is in `certs/closedform_dd_qq.jsonl`.

Proof. For each N , `scripts/closed_form.py` evaluates t_{cf} in Arb ball arithmetic at 256 bits from the rational enclosure of τ_N (Lemma 19) and forms $(t_{\text{cf}} - t^*)/t^*$ with the integer t^* of Theorem 12; the bounds stated are the extremes over N of the resulting enclosures, rounded outwards. The same quantities were evaluated with `mpmath.iv`; the two enclosures overlap for every N . The values of N with a tie at the maximum ($N \leq 4$) are below the range of the statement. $\square$

Remark 22. In two dimensions (both values of q) the relative error $(t_{\text{cf}} - t^*)/t^*$ is positive for $N \leq 13$ and negative for every $N \geq 14$ of the certified ranges; its modulus has a flat maximum of about 0.97% near $N = 137$ ($q = 4/5$) and decreases slowly beyond, though not monotonically (Table 5). In three dimensions it is positive for every $3 \leq N \leq 60$ at $q = 4/5$ (negative only at $N = 2$) and for every $N \neq 3$ of the range at $q = 1/2$; at $q = 4/5$ it falls from 0.00275 at $N = 10$ to 0.000144 at $N = 60$, but not monotonically. The integer rounding of the mode contributes up to $1/t^*$ to these relative errors; in three dimensions this is comparable with the error itself at the upper end of the range, and it is the reason why the error is not monotone there. For $2 \leq N \leq 9$ the closed form is poorer (up to 32% at $N = 2$ in two dimensions, $q = 4/5$); these cases are recorded in `certs/closedform_summary.json`. The constants ε_d are statements about the stated finite ranges only; that they persist for all $N \geq 10$ is Conjecture 48.

6 Two constants

6.1 The one-dimensional constant

For $\tau > 0$ let

$$\varphi(\tau) = \sum_{m \geq 1} (-1)^{m+1} (2m-1) e^{-(2m-1)^2 \pi^2 \tau / 4}.$$

($\pi\varphi(Dt/L^2) D/L^2$ is the first-passage density at the absorbing end of an interval of length L for Brownian motion with diffusivity D started at the reflecting end; its mean is $L^2/2D$, so the continuum ratio mode/mean equals $2\tau^*$ with τ^* the maximiser of φ .)

d	q	range of N	bound on $ t_{\text{cf}} - t^* /t^*$	worst N	interval containing $(t_{\text{cf}} - t^*)/t^*$
1	1/2	$10 \leq N \leq 300$	0.228187	10	[0.159008, 0.228187]
1	1/2	$20 \leq N \leq 300$	0.184401	21	[0.159008, 0.184401]
1	1/2	$35 \leq N \leq 300$	0.171773	36	[0.159008, 0.171773]
1	1/2	$50 \leq N \leq 300$	0.167141	51	[0.159008, 0.167141]
1	1/2	$100 \leq N \leq 300$	0.162171	102	[0.159008, 0.162171]
1	4/5	$10 \leq N \leq 600$	0.225045	11	[0.158243, 0.225045]
1	4/5	$20 \leq N \leq 600$	0.186954	21	[0.158243, 0.186954]
1	4/5	$35 \leq N \leq 600$	0.172733	35	[0.158243, 0.172733]
1	4/5	$50 \leq N \leq 600$	0.167595	50	[0.158243, 0.167595]
1	4/5	$100 \leq N \leq 600$	0.162246	101	[0.158243, 0.162246]
1	4/5	$N = 800$	0.158049	800	[0.158048, 0.158049]
1	4/5	$N = 1000$	0.157936	1000	[0.157935, 0.157936]
2	1	$10 \leq N \leq 80$	0.009523	76	[-0.009523, 0.005467]
2	1	$20 \leq N \leq 80$	0.009523	76	[-0.009523, -0.003201]
2	1	$35 \leq N \leq 80$	0.009523	76	[-0.009523, -0.007698]
2	1	$50 \leq N \leq 80$	0.009523	76	[-0.009523, -0.008815]
2	1/2	$10 \leq N \leq 80$	0.009540	80	[-0.009540, 0.006710]
2	1/2	$20 \leq N \leq 80$	0.009540	80	[-0.009540, -0.004819]
2	1/2	$35 \leq N \leq 80$	0.009540	80	[-0.009540, -0.007947]
2	1/2	$50 \leq N \leq 80$	0.009540	80	[-0.009540, -0.008933]
2	4/5	$10 \leq N \leq 160$	0.009698	137	[-0.009698, 0.005978]
2	4/5	$20 \leq N \leq 160$	0.009698	137	[-0.009698, -0.004006]
2	4/5	$35 \leq N \leq 160$	0.009698	137	[-0.009698, -0.007897]
2	4/5	$50 \leq N \leq 160$	0.009698	137	[-0.009698, -0.008910]
2	4/5	$100 \leq N \leq 160$	0.009698	137	[-0.009698, -0.009606]
2	4/5	$N = 180$	0.009661	180	[-0.009661, -0.009660]
2	4/5	$N = 200$	0.009639	200	[-0.009639, -0.009638]
3	1	$10 \leq N \leq 32$	0.004534	10	[0.000383, 0.004534]
3	1	$20 \leq N \leq 32$	0.001173	21	[0.000383, 0.001173]
3	1/2	$10 \leq N \leq 32$	0.001562	10	[0.000177, 0.001562]
3	1/2	$20 \leq N \leq 32$	0.000386	20	[0.000177, 0.000386]
3	4/5	$10 \leq N \leq 60$	0.002749	10	[0.000116, 0.002749]
3	4/5	$20 \leq N \leq 60$	0.000945	21	[0.000116, 0.000945]
3	4/5	$35 \leq N \leq 60$	0.000249	37	[0.000116, 0.000249]
3	4/5	$50 \leq N \leq 60$	0.000195	50	[0.000116, 0.000195]

Table 5: Certified bounds for the relative error of the closed form on sub-ranges. The last column is an interval containing $(t_{\text{cf}} - t^*)/t^*$ for every N of the range. For $q = 1$ the mode is the raw (parity-affected) argmax.

Theorem 23 (the constant $\kappa_1 = 2\tau^*$). φ has exactly one critical point τ^* in $(0, \infty)$; φ is strictly increasing on $(0, \tau^*]$ and strictly decreasing on $[\tau^*, \infty)$; and

$$0.16664213274747394937426345812 < \tau^* < 0.16664213274747394937426345843,$$

$$\kappa_1 := 2\tau^* \in (0.33328426549494789874852691624, 0.33328426549494789874852691686).$$

Proof. Step 1 (dual series). For all $\tau > 0$,

$$\varphi(\tau) = (\pi\tau)^{-3/2} \sum_{n \geq 0} (-1)^n (2n+1) e^{-(2n+1)^2/(4\tau)}. \quad (5.1)$$

Indeed, by Jacobi's identity $\prod_{k \geq 1} (1 - x^k)^3 = \sum_{n \geq 0} (-1)^n (2n+1) x^{n(n+1)/2}$ (Hardy and Wright, *An Introduction to the Theory of Numbers*, Theorem 357), the Dedekind function $\eta(z) = e^{\pi iz/12} \prod_{k \geq 1} (1 - e^{2\pi i k z})$ satisfies $\eta(z)^3 = \sum_{n \geq 0} (-1)^n (2n+1) e^{\pi iz(2n+1)^2/4}$, so $\varphi(\tau) = \eta(i\pi\tau)^3$; and $\eta(-1/z) = (-iz)^{1/2} \eta(z)$ (Apostol, *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed., Theorem 3.1) with $z = i/(\pi\tau)$ gives $\eta(i\pi\tau)^3 = (\pi\tau)^{-3/2} \eta(i/(\pi\tau))^3$, which is (5.1). (As a safeguard against a mis-transcribed normalisation, both sides of (5.1) were evaluated in ball arithmetic at six values of τ between $1/20$ and 2 ; the enclosures overlap to 30 digits.)

Step 2 (large τ). Differentiating termwise, $-\frac{4}{\pi^2} \varphi'(\tau) = \sum_{m \geq 1} (-1)^{m+1} a_m$ with $a_m = (2m-1)^3 e^{-(2m-1)^2 \pi^2 \tau/4} > 0$ and $a_{m+1}/a_m = \left(\frac{2m+1}{2m-1}\right)^3 e^{-2m\pi^2 \tau} \leq 27e^{-2\pi^2 \tau}$. For $\tau > \tau_1 := \ln 27/(2\pi^2)$ the a_m decrease strictly to 0, so the alternating sum exceeds $a_1 - a_2 > 0$. Hence $\varphi' < 0$ on (τ_1, ∞) , and $\tau_1 < 0.16697 < 1/4$.

Step 3 (small τ). Differentiating (5.1) termwise, for $\tau > 0$,

$$\pi^{3/2} \tau^{7/2} e^{1/(4\tau)} \varphi'(\tau) = B(\tau) := \frac{1}{4} - \frac{3\tau}{2} - E(\tau), \quad E(\tau) = \sum_{n \geq 1} (-1)^{n+1} b_n(\tau),$$

$$b_n(\tau) = (2n+1) \left(\frac{(2n+1)^2}{4} - \frac{3\tau}{2} \right) e^{-n(n+1)/\tau}.$$

Let $0 < \tau \leq 1/4$. Then $b_n > 0$ and $\frac{b_{n+1}}{b_n} = \frac{2n+3}{2n+1} \cdot \frac{(2n+3)^2 - 6\tau}{(2n+1)^2 - 6\tau} e^{-2(n+1)/\tau} \leq \frac{5}{3} \cdot \frac{47}{15} e^{-16} < 1$, so the b_n decrease to 0 and

$$b_1 - b_2 \leq E \leq b_1 - b_2 + b_3. \quad (5.2)$$

Moreover $|b'_n(\tau)| \leq (2n+1) e^{-n(n+1)/\tau} \left[\frac{3}{2} + \frac{(2n+1)^2}{4} \frac{n(n+1)}{\tau^2} \right] \leq \frac{441}{2} n^5 e^{-4n(n+1)}$, because $u \mapsto u^2 e^{-n(n+1)u}$ is decreasing for $u = 1/\tau \geq 4$ and $(2n+1) \left[\frac{3}{2} + 4(2n+1)^2 n(n+1) \right] \leq 3n \cdot \frac{147}{2} n^4$. Hence $\sum_n |b'_n| \leq \frac{441}{2} (e^{-8} + \frac{32e^{-24}}{1-e^{-11}}) < 0.074$, the series for E may be differentiated termwise, and

$$B'(\tau) = -\frac{3}{2} - E'(\tau) < -\frac{3}{2} + 0.074 < 0 \quad (0 < \tau \leq \frac{1}{4}).$$

So B is strictly decreasing on $(0, 1/4]$, with $B(0+) = 1/4 > 0$ and $B(1/4) \leq \frac{1}{4} - \frac{3}{8} < 0$ (as $E \geq 0$). It has exactly one zero $\tau^* \in (0, 1/4)$, and φ' has the sign of B there.

Step 4. By Steps 2 and 3, $\varphi' > 0$ on $(0, \tau^*)$, $\varphi' < 0$ on $(\tau^*, 1/4]$ and on $(\tau_1, \infty) \supset [1/4, \infty)$.

Step 5 (enclosure). `scripts/constants.py` bisection on rational points, using (5.2): at the two rationals $\tau_- < \tau_+$ recorded in `certs/constants.json`, $\frac{1}{4} - \frac{3}{2}\tau_- - (b_1 - b_2 + b_3)(\tau_-) > 0$ and $\frac{1}{4} - \frac{3}{2}\tau_+ - (b_1 - b_2)(\tau_+) < 0$ in Arb ball arithmetic at 512 bits, and again in `mpmath.iv`. Hence $\tau_- < \tau^* < \tau_+$; the decimal bounds in the statement are these rationals rounded outwards. $\square$

Remark 24. The single-image approximation is the zero $1/6$ of $\frac{1}{4} - \frac{3}{2}\tau$, i.e. the ratio $1/3$; the correction $E(\tau^*) \approx b_1(1/6) = 6e^{-12}$ moves the ratio down by 4.9×10^{-5} . The value quoted in Section 5 of the article, 0.3332842655, is correct to all printed digits.

6.2 The two-dimensional constant

Theorem 25 (value of C_2). *Let $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi})$ (the lemniscate constant) and*

$$C_2 := \frac{8}{\pi} \left[\gamma + \ln \frac{4\sqrt{2}}{\varpi} - \frac{1}{2} - \frac{\pi}{4} \right] = \frac{8}{\pi} \left[\gamma + 4 \ln 2 + \frac{1}{2} \ln \pi - 2 \ln \Gamma\left(\frac{1}{4}\right) \right] - 2 - \frac{4}{\pi}.$$

Then $|C_2 - 0.15463778781891726249073383080028304919172661983273| < 10^{-50}$.

Proof. The two expressions are equal because $\ln(4\sqrt{2}/\varpi) = \ln(16\sqrt{\pi}/\Gamma(1/4)^2)$. The enclosure is the Arb evaluation at 512 bits (radius $< 3 \times 10^{-71}$) in `scripts/constants.py`; the `mpmath.iv` evaluation of the first expression gives an interval of width $< 10^{-150}$ which overlaps it. $\square$

Theorem 25 certifies the *number*. That C_2 is the coefficient of N^2 in the expansion of $q \mathbb{E}T$ in two dimensions is Theorem 4.5 of the article; that note states that its remainder estimate is only sketched. The following finite-range statement does not depend on any asymptotic argument.

Theorem 26 (two-term MFPT law in two dimensions on a finite range; computer-assisted). *Let $d = 2$, $\tau_N = q \mathbb{E}_{x_0} T$ and $r_N := \tau_N - \frac{8}{\pi} N^2 \ln N - C_2 N^2$. For every $2 \leq N \leq 301$,*

$$0.3211 < r_N < 0.53213,$$

and r_N is strictly increasing in N on this range ($r_2 = 0.32110964\dots$, $r_{10} = 0.52165509\dots$, $r_{100} = 0.53203394\dots$, $r_{200} = 0.53211414\dots$, $r_{300} = 0.53212899\dots$).

Proof. `scripts/mfpt2d_remainder.py`: Arb evaluation of r_N from the rational enclosures of τ_N of Lemma 19; monotonicity is the certified inequality $\sup r_N < \inf r_{N+1}$ for consecutive N . (The enclosures of Lemma 19 were also computed for five isolated larger values; they give, unconditionally, $r_{350} = 0.53213215\dots$, $r_{401} = 0.53213423\dots$, $r_{450} = 0.53213560\dots$, $r_{500} = 0.53213660\dots$, $r_{600} = 0.53213791\dots$, all below C_0 of Proposition 27.) $\square$

Proposition 27 (extension, conditional on the single-sum formula). *Assume the single-sum formula $\tau_N = 4N \sum_{k=1}^{N-1} c_k^2 \frac{\sqrt{1+s_k^2}}{s_k} g_k$ with $s_k = \sin \frac{\pi k}{2N}$, $c_k = \cos \frac{\pi k}{2N}$, $g_k = \coth(N \operatorname{arsinh} s_k)$ for odd k and $\tanh(N \operatorname{arsinh} s_k)$ for even k (Theorem 2(b) of the companion note, where it is proved). Then $r_N < 0.5321405 < C_0 := \varpi^4/(9\pi^2) = 0.53214088\dots$ for $301 < N \leq 1500$, and $r_N < r_{N+1}$ for $301 \leq N < 1500$; together with Theorem 26, r_N is strictly increasing on the whole range $2 \leq N \leq 1500$.*

Proof. Arb evaluation of the single sum, same script; monotonicity for $301 < N < 1500$ is the certified inequality $\sup r_N < \inf r_{N+1}$, as in Theorem 26. At the junction of the two ranges, the upper bound 0.532129078393 of r_{301} (from the linear-system enclosure, Theorem 26) is below the lower bound 0.532129156439 of the next value of r_N (from the single sum); both decimals are rounded outwards (`scripts/check_remarks.py`, and again with `mpmath.iv` for r_{301} in `scripts/audit.py`). As a consistency check, the single sum was also evaluated for all $N \leq 60$, every twentieth $N \leq 301$ and $N = 350, 401, 450, 500, 600$; it overlaps the linear-system enclosure in every case. $\square$

7 A theorem for all N : log-concavity in one dimension for $q \leq 1/2$

Theorem 28. *Let $d = 1$, $N \geq 2$ and $0 < q \leq 1/2$ (real). Then $f(t) > 0$ exactly for $t \geq N - 1$, and*

$$f(t)^2 \geq f(t-1)f(t+1) \quad \text{for all } t \geq 1.$$

Consequently there is $t^ \geq N - 1$ with $f(t) < f(t+1)$ for $N - 2 \leq t < t^*$ and $f(t) \geq f(t+1)$ for $t \geq t^*$: the PMF is unimodal (possibly with a plateau of equal values at the top).*

Proof. Here $\Omega' = \{0, \dots, N-2\}$, $n := N-1$ sites, Q is symmetric tridiagonal with off-diagonal entries $q/2$, and $f(t) = \frac{q}{2}(Q^{t-1})_{n-1,0}$.

Step 1 (generating function). By Lemma 2 the spectral radius of Q is less than 1, so for $|z| \leq 1$ the matrix $I - zQ$ is invertible and $F(z) := \sum_{t \geq 1} f(t)z^t = \frac{qz}{2}[(I - zQ)^{-1}]_{n-1,0}$. By Cramer's rule this entry is $(-1)^{n-1} \det(\text{minor}) / \det(I - zQ)$, where the minor is obtained from $I - zQ$ by deleting row 0 and column $n-1$; that minor is upper triangular with diagonal entries $-zq/2$. Hence

$$F(z) = \frac{(qz/2)^n}{\det(I - zQ)} = \frac{(qz/2)^n}{\prod_{j=1}^n (1 - \theta_j z)},$$

with $\theta_1, \dots, \theta_n$ the (real) eigenvalues of Q . Since $F(1) = \mathbb{P}(T < \infty) = 1$ (Lemma 2), $(q/2)^n = \prod_j (1 - \theta_j)$, and

$$F(z) = \prod_{j=1}^n \frac{(1 - \theta_j)z}{1 - \theta_j z}. \quad (6.1)$$

Step 2 (eigenvalues). $Q = (1 - q)I + qQ_1$ with Q_1 symmetric and $\|Q_1\|_\infty \leq 1$, so $\theta_j \geq 1 - 2q \geq 0$; also $\theta_j < 1$ by Lemma 2, hence $\theta_j \in [0, 1)$. As $Q \neq 0$ is symmetric, at least one θ_j is positive; let it be θ_1 .

Step 3 (convolution). For $\theta \in [0, 1)$ the factor $(1 - \theta)z/(1 - \theta z)$ is the generating function of $g_\theta(k) = (1 - \theta)\theta^{k-1}$, $k \geq 1$ (for $\theta = 0$: of δ_1). By (6.1), $f = g_{\theta_1} * g_{\theta_2} * \dots * g_{\theta_n}$.

Step 4 (one convolution preserves log-concavity). Claim: let $g : \mathbb{Z} \rightarrow [0, \infty)$ with $g(t) > 0 \iff t \geq m$ and $g(t)^2 \geq g(t-1)g(t+1)$ for all t , let $\theta \in [0, 1)$ and $h = g * g_\theta$. Then $h(t) > 0 \iff t \geq m+1$ and $h(t)^2 \geq h(t-1)h(t+1)$ for all t . If $\theta = 0$, $h(t) = g(t-1)$. Let $\theta > 0$. The support statement is clear. From the definition, $h(t+1) = \theta h(t) + (1 - \theta)g(t)$, so for $t \geq m+1$

$$\frac{h(t+1)}{h(t)} = \theta + (1 - \theta) \frac{g(t)}{h(t)}, \quad \frac{h(t)}{g(t)} = \sum_{k=1}^{t-m} (1 - \theta)\theta^{k-1} \prod_{i=1}^k \frac{1}{\rho(t-i)}, \quad \rho(u) := \frac{g(u+1)}{g(u)}.$$

By hypothesis ρ is non-increasing on $u \geq m$. When t is replaced by $t+1$, every factor $1/\rho(t-i)$ is replaced by $1/\rho(t+1-i) \geq 1/\rho(t-i)$ and one more non-negative term is added, so $h(t)/g(t)$ is non-decreasing in $t \geq m+1$. Therefore $h(t+1)/h(t)$ is non-increasing on $t \geq m+1$, which is $h(t)^2 \geq h(t-1)h(t+1)$ for $t \geq m+2$; for $t \leq m+1$ the right-hand side vanishes.

Step 5. g_{θ_1} satisfies the hypotheses of the claim with $m = 1$ (its ratio is constant). Applying the claim $n-1$ times gives the theorem, with support $\{t \geq n\}$. The unimodality statement follows because $f(t+1)/f(t)$ is non-increasing on $t \geq N-1$ and cannot stay above 1 (as $\sum f = 1$). $\square$

Remark 29. The hypothesis $q \leq 1/2$ cannot be replaced by $q < 1$. For $N = 3$ this can be seen exactly. Here $\Omega' = \{0, 1\}$ and Q has the rows $(1 - \frac{q}{2}, \frac{q}{2})$ and $(\frac{q}{2}, 1 - q)$; its eigenvalues $x \neq y$ are distinct because the off-diagonal entry is not zero, and by the Cayley–Hamilton theorem $Q^k = \alpha_k Q + \beta_k I$ with $\alpha_k = (x^k - y^k)/(x - y)$. Hence $f(t) = \frac{q}{2}(Q^{t-1})_{1,0} = \frac{q^2}{4} \alpha_{t-1}$, and the identity $(x^k - y^k)^2 - (x^{k-1} - y^{k-1})(x^{k+1} - y^{k+1}) = (xy)^{k-1}(x - y)^2$ gives

$$f(t)^2 - f(t-1)f(t+1) = \frac{q^4}{16} (\det Q)^{t-2} \quad (t \geq 2), \quad \det Q = 1 - \frac{3q}{2} + \frac{q^2}{4}.$$

So for $d = 1$, $N = 3$ the PMF is log-concave if and only if $\det Q \geq 0$, i.e. $q \leq 3 - \sqrt{5} = 0.7639\dots$; for $3 - \sqrt{5} < q \leq 1$ log-concavity fails at every odd $t \geq 3$ and at no other t (at $q = 4/5$, $\det Q = -1/25$). For $N = 4$ and $q = 4/5$ it fails at $t = 4$, and exact rational arithmetic finds no other failure for $t \leq 400$. Unimodality nevertheless holds for $q = 4/5$ in the whole certified range (Theorem 12). Formula (6.1) is the discrete-time form of the classical Karlin–McGregor/Keilson representation of birth-and-death passage times; it is proved above in two lines for the present chain, so no citation is needed.

Theorem 30 (the one-dimensional mode on a finite range; computer-assisted). *Let $d = 1$, $\kappa_1 = 2\tau^*$ as in Theorem 23, and $y_N(q) = \kappa_1(N - \frac{1}{2})^2/q$.*

- (a) *For $q = 4/5$ and every $10 \leq N \leq 600$: $-1.097 < t^* - y_N < -0.099$.*
- (b) *For $q = 1/2$ and every $10 \leq N \leq 300$: $-1.160 < t^* - y_N < -0.164$.*
- (c) *In both cases $|t^*/\mathbb{E}T - \kappa_1| \leq c/(N(N-1))$ with $c = 0.80$ for $q = 4/5$ and $c = 0.50$ for $q = 1/2$ (recall $\mathbb{E}T = N(N-1)/q$).*

In (a) and (b) the interval has length less than 1, so t^ is the unique integer in it: in the certified range the mode is given exactly by $t^* = \lfloor y_N - 0.099 \rfloor$ for $q = 4/5$ and by $t^* = \lfloor y_N - 0.164 \rfloor$ for $q = 1/2$.*

Proof. `scripts/closed_form.py` forms $t^* - \kappa_1(N - \frac{1}{2})^2/q$ in exact rational arithmetic from the certified integer t^* and the rational enclosure of κ_1 of Theorem 23; `scripts/summarise.py` takes the extremes over N , and `scripts/audit.py` re-checks (a)–(c) and the floor formula for every N . $\square$

Remark 31. Theorem 30 is a finite-range statement. That $t^*/\mathbb{E}T \rightarrow \kappa_1$ as $N \rightarrow \infty$ is not proved here (Section 9); the theorem shows that in the certified range the lattice ratio differs from the continuum constant by less than N^{-2} . The two isolated values $N = 800, 1000$ at $q = 4/5$ also satisfy (a) (`certs/SUMMARY.json`, key `law_1d`).

Proposition 32 (one dimension, q close to 1: not unimodal, for all $N \geq 4$). *Let $d = 1$, $N \geq 4$ and $q_N := \frac{2N-4}{2N-3}$. For every real q with $q_N \leq q \leq 1$,*

$$f(N+1) > f(N), \quad \text{and} \quad f(N-1) > f(N) \iff q > q_N.$$

Consequently, for $q_N < q \leq 1$ the PMF satisfies $f(N-1) > f(N) < f(N+1)$ and is not unimodal, even in the weak sense of Definition 3. In particular, for $q = 1$ the one-dimensional PMF is not unimodal for every $N \geq 4$.

Proof. A step moves the walker by at most one site, so a path from 0 that reaches the target $N-1$ for the first time at step $N-1+k$ consists of $N-1$ forward moves and k further steps. Put $n = N-2$, $u = 1-q$, $m = (q/2)^{N-1}$, and let $r_0 = 1 - \frac{q}{2} = \frac{1+u}{2}$ be the probability of staying at the reflecting end 0 (rest or cancelled move) and $r = 1-q = u$ the probability of resting at one of the n sites $1, \dots, N-2$.

$k=0$: only the direct path, $f(N-1) = m$. $k=1$: the direct path with one stay inserted at one of the $N-1$ non-target sites (a backward move would cost two extra steps), so $f(N) = m(r_0 + nr)$. Hence $f(N-1) > f(N)$ iff $r_0 + nr < 1$ iff $n(1-q) < \frac{q}{2}$ iff $q > q_N$. $k=2$: either two stays (at the same or at different sites), with total weight $m(r_0^2 + nr_0r + \frac{n(n+1)}{2}r^2)$, or one backward move from a site $j \in \{1, \dots, N-2\}$ to $j-1$ followed by one more forward move, with total weight $mn(q/2)^2$ (at the site 0 a backward move is cancelled and has been counted as a stay). Therefore $f(N+1) - f(N) = mG(u)$ with

$$G(u) = \frac{(1+u)^2}{4} + \frac{nu(1+u)}{2} + \frac{n(n+1)}{2}u^2 + \frac{n(1-u)^2}{4} - \frac{1+u}{2} - nu = \frac{n-1}{4} - nu + \frac{2n^2+5n+1}{4}u^2.$$

The condition $q_N \leq q \leq 1$ is $0 \leq u \leq \frac{1}{2n+1}$. The quadratic G is decreasing on $[0, u_0]$, $u_0 = \frac{2n}{2n^2+5n+1}$, and $u_0 \geq \frac{1}{2n+1}$ because $2n(2n+1) - (2n^2+5n+1) = 2n^2-3n-1 \geq 1$ for $n \geq 2$. So on the interval

$$G(u) \geq G\left(\frac{1}{2n+1}\right) = \frac{(n-1)(2n+1)^2 - 4n(2n+1) + 2n^2 + 5n + 1}{4(2n+1)^2} = \frac{n(2n^2-3n-1)}{2(2n+1)^2} > 0.$$

Finally, if $f(N-1) > f(N) < f(N+1)$ then no t^* as in Definition 3 exists: $f(N-1) > f(N)$ forces $t^* \leq N-1$, and $f(N) < f(N+1)$ forces $t^* \geq N+1$. $\square$

Remark 33. At $q = q_N$ the proposition gives $f(N-1) = f(N) < f(N+1)$; for $N = 4$ this is the exceptional case $(1, \frac{4}{5}, 4)$ of Theorem 12. Proposition 32 shows that the restriction on q in Conjecture 46 cannot be removed: in one dimension unimodality fails for every $q > (2N-4)/(2N-3)$, a threshold which equals $4/5$ for $N = 4$ and increases to 1. Since $q_N \geq 4/5$ for all $N \geq 4$, the value $q = 4/5$ of the article is the largest activity for which one-dimensional unimodality for all N is not excluded by this mechanism. For $q = 1$ the proposition is consistent with the certificates of Theorems 13 and 40 (at least two strict local maxima for all $4 \leq N \leq 400$).

8 A second engine and extended ranges

The ranges of Table 1 were limited by the speed of the Python-integer checker. This section describes a second engine written in C with `unsigned __int128` integers (`scripts/c128/fp128.c`, 366 lines including comments). In the wall-clock times recorded on the same (heavily loaded) machine for the upper parts of the ranges of Table 1, it was between 4.7 and 71.9 times faster than the generator (medians 8.7 to 44.7, depending on d and q) and between 11.1 and 270.1 times faster than the Python-integer checker (medians 17.2 to 206.7); these ratios of wall-clock times are indicative only. The section gives the lemmas that make its output a proof, the tests that validate it against the programs of Section 3, and the theorems obtained with it on larger ranges of N . Its trusted base is different from (T1), so its results are stated separately (Theorems 39 and 40); on the ranges of Table 1 it reproduces every verdict of Theorems 12 and 13.

Trusted base (T5). C99 unsigned integer arithmetic on `uint64_t` and `unsigned __int128` (+, −, ×, /, %, shifts, bitwise and/or, comparisons), as compiled by Apple clang 21.0.0 for arm64 with `-O2`; by Lemma 36 no operation wraps around. The program uses no floating-point arithmetic. The verdicts are formed from its output by `scripts/c128/c128_post.py` with Python integers (T1), which also re-checks the tail inequality site by site.

8.1 The engine

The engine differs from the programs of Section 3 in three respects.

- (i) *No exact phase.* The recurrences of Lemma 7 are started at $t_s = 1$ with $L_1 = U_1 = 2^P e_{x_0}$, $P = 112$ (the lemma allows any $t_s \geq 1$; here $2^P X_1/D^0$ is an integer vector). This gives enclosures $F_t^- \leq 2^P \sigma_t \leq F_t^+$ for every $t \geq 1$, which are useless for comparing σ_t with σ_{t+1} as long as $\sigma_t \ll 2^{-P}$.
- (ii) *A floating-interval side computation for the early relations.* While $F_t^- < 2^{48}$, a second pair of vectors $\text{lo}_t \leq v_t \leq \text{hi}_t$ is carried in a software floating format with 63-bit integer mantissas and directed rounding (Lemma 37). Its *relative* width at every site grows by at most about 2^{-58} per step, whatever the size of $v_t(y)$ (in one application of `fsum` of Lemma 37 the truncations of the terms cost less than $\sum_j c_j \leq 15$ units of $\text{acc} \geq 2^{62}$, and the final rounding less than one unit of the mantissa $m \geq 2^{62}$; this estimate is not used in any proof), and it supplies certified relations between σ_t and σ_{t+1} in the early phase whenever their relative difference exceeds the relative width of the two enclosures.
- (iii) *Symmetry reduction.* Only the sites with $y_1 \leq y_2 \leq \dots \leq y_d$ are stored (Lemma 35); this divides the work by almost $d!$.

Lemma 34 (time of the first possible passage). *Let $d \geq 1$, $N \geq 2$ and $q \in (0, 1]$ be real. For $t \geq 1$ and $y \in \Omega'$: $v_t(y) > 0$ if and only if $t-1 \geq |y|_1$. Consequently $\sigma_t > 0$ if and only if $t \geq d(N-1)$, that is, $f(t) = 0$ for $t < d(N-1)$ and $f(t) > 0$ for all $t \geq d(N-1)$.*

Proof. $v_t(y) = (Q^{t-1})(y, x_0)$ is the sum, over all sequences $x_0 = z_0, z_1, \dots, z_{t-1} = y$ in Ω' , of the products $\prod_i Q(z_{i+1}, z_i) \geq 0$. A product is positive only if consecutive sites are equal or lattice neighbours, so $|z_{i+1}|_1 \leq |z_i|_1 + 1$ and $t - 1 \geq |y|_1$ is necessary. Conversely, $Q(x_0, x_0) = 1 - \frac{d}{2} > 0$ (the corner x_0 has d cancelled moves), and there is a coordinatewise increasing lattice path of length $|y|_1$ from x_0 to y ; it stays in Ω' because its sites z satisfy $z \leq y$ coordinatewise and $y \neq a$. Staying $t - 1 - |y|_1$ times at x_0 and then following this path gives a positive product. Finally the d neighbours of a have $|y|_1 = d(N - 1) - 1$. $\square$

Lemma 35 (symmetry reduction). *Let the symmetric group S_d act on Ω by permuting coordinates; call a function Z on Ω symmetric if $Z(\pi y) = Z(y)$ for all $\pi \in S_d$, $y \in \Omega$. For $y \in \Omega'$ let $\nu_1(y), \dots, \nu_{2d}(y)$ be the list of the sites $y \pm e_k$, with $y \pm e_k$ replaced by y itself when it lies outside Ω (a cancelled move).*

- (a) $\pi x_0 = x_0$, $\pi a = a$, and the list $\nu_1(\pi y), \dots, \nu_{2d}(\pi y)$ is a rearrangement of $\pi \nu_1(y), \dots, \pi \nu_{2d}(y)$.
- (b) For every function Z on Ω with $Z(a) = 0$: $(MZ)(y) = c_s Z(y) + c_m \sum_{j=1}^{2d} Z(\nu_j(y))$ for $y \in \Omega'$.
- (c) Let Ψ be any function of a number and an unordered list of $2d$ numbers, and for Z with $Z(a) = 0$ define $(\hat{\Psi}Z)(y) = \Psi(Z(y); \{Z(\nu_j(y))\}_j)$ for $y \in \Omega'$ and $(\hat{\Psi}Z)(a) = 0$. If Z is symmetric, so is $\hat{\Psi}Z$. Consequently X_t, v_t , the vectors L_t, U_t of Lemma 7 with $t_s = 1$, and the vectors lo_t, hi_t of Lemma 37 are symmetric for every $t \geq 1$.
- (d) Let $\Omega_{\leq} = \{y \in \Omega : y_1 \leq \dots \leq y_d\}$ and let $\text{sort}(z) \in \Omega_{\leq}$ be the non-decreasing rearrangement of z . A symmetric Z is determined by its restriction to $\Omega_{\leq}$, $Z(z) = Z(\text{sort}(z))$; an inequality between two symmetric functions holds on Ω' if and only if it holds on $\Omega_{\leq} \setminus \{a\}$; and $\sum_{y \sim a} Z(y) = d Z((N - 2, N - 1, \dots, N - 1))$.

Proof. (a) A permutation of the coordinates maps Ω onto itself, fixes the two constant vectors x_0 and a , and maps the set of the $2d$ unit vectors $\pm e_k$ onto itself; $y + e \notin \Omega$ if and only if $\pi y + \pi e \notin \Omega$. (b) is the definition of $M = c_m(A + B) + c_s I$ on Ω' : a neighbour in Ω' contributes $c_m Z$ of that neighbour, the neighbour a contributes $0 = c_m Z(a)$, and each of the $b(y)$ cancelled moves contributes $c_m Z(y)$. (c) By (a), $\{Z(\nu_j(\pi y))\}_j = \{Z(\pi \nu_j(y))\}_j = \{Z(\nu_j(y))\}_j$ as unordered lists, and $Z(\pi y) = Z(y)$. The vectors named are obtained from the symmetric vector e_{x_0} (times a constant) by iterating maps of the form $\hat{\Psi}$: by (b), $\Psi = c_s z + c_m \sum_j z_j$ for X_t , the same divided by D for v_t , followed by $\lfloor \cdot / D \rfloor$ or $\lceil \cdot / D \rceil$ for L_t, U_t , and the procedure `fsum` of Lemma 37 for lo_t, hi_t (its result does not depend on the order of the terms: E is a maximum and acc a sum of integers). (d) Every orbit of S_d meets $\Omega_{\leq}$ in exactly one point; the d neighbours $a - e_k$ of a form one orbit. $\square$

Lemma 36 (integer routines of the C engine). *Let $1 \leq D \leq 15$ and $P = 112$.*

- (a) Let $0 \leq W < 2^{116}$ be an integer, $w_1 = \lfloor W/2^{58} \rfloor$, $w_0 = W - 2^{58}w_1$, $q_1 = \lfloor w_1/D \rfloor$, $r_1 = w_1 - Dq_1$. Then $w_1 < 2^{58}$, $c := 2^{58}r_1 + w_0 < 2^{62}$ and $\lfloor W/D \rfloor = 2^{58}q_1 + \lfloor c/D \rfloor$ (routine `divfloor`; all divisions are 64-bit).
- (b) Let $0 \leq W < 2^{126}$ and $1 \leq D' \leq 15$. Writing $W = 2^{84}a_2 + 2^{42}a_1 + a_0$ with $0 \leq a_1, a_0 < 2^{42}$ (so $a_2 < 2^{42}$), three steps of long division in base 2^{42} , $a_2 = D'q_2 + r_2$, $2^{42}r_2 + a_1 = D'q_1 + r_1$, $2^{42}r_1 + a_0 = D'q_0 + r_0$ ($0 \leq r_i < D'$), give $\lfloor W/D' \rfloor = 2^{84}q_2 + 2^{42}q_1 + q_0$, and every intermediate number is less than 2^{46} (routine `divsmall`).
- (c) If $0 \leq L \leq U \leq 2^P$ entrywise and $L(a) = U(a) = 0$, the routine `step_fixed` returns $\lfloor ML/D \rfloor$ and $\lceil MU/D \rceil$, and no 128-bit operation wraps around: every sum of $2d$ entries is at most $2d2^P$, $W = c_m(\text{sum}) + c_s(\text{entry}) \leq D2^P < 2^{116}$, and $W + D - 1 < 2^{116}$.

Proof. (a) $W = D 2^{58} q_1 + c$ with $0 \leq c < 2^{58}(r_1 + 1) \leq 2^{58}D < 2^{62}$, hence $\lfloor W/D \rfloor = 2^{58}q_1 + \lfloor c/D \rfloor$. (b) is the schoolbook division: $W = D'(2^{84}q_2 + 2^{42}q_1 + q_0) + r_0$, and $2^{42}r_i + a_{i-1} < 2^{42}D' < 2^{46}$. (c) By Lemma 35(b) the routine forms $W = (MZ)(y)$; the bounds follow from $2dc_m + c_s = D$ and $D 2^{112} + D - 1 < 2^{116}$; $\lceil W/D \rceil = \lfloor (W + D - 1)/D \rfloor$ for integers; (a) applies. Lemma 7 gives $U_{t+1} \leq 2^P$, so the hypothesis propagates. $\square$

Lemma 37 (floating enclosures). *Let $\mathbb{F} = \{0\} \cup \{m 2^e : m, e \in \mathbb{Z}, 2^{62} \leq m < 2^{63}\}$.*

(a) (*fsum*) *Let $x_1, \dots, x_k \in \mathbb{F}$, let $c_1, \dots, c_k \geq 0$ be integers with $\sum_j c_j \leq 15$, let $1 \leq D' \leq 15$ and $s = \frac{1}{D'} \sum_j c_j x_j$. The routine *fsum* returns $z_\downarrow \in \mathbb{F}$ with $z_\downarrow \leq s$ (mode “down”) or $z_\uparrow \in \mathbb{F}$ with $z_\uparrow \geq s$ (mode “up”); the result is 0 if and only if $s = 0$; no operation wraps around; and the exponent of a non-zero result lies in $[E - 4, E + 5]$, where E is the largest exponent of a non-zero term.*

(b) *Define $\text{lo}_1 = \text{hi}_1 = e_{x_0}$ and, for $y \in \Omega'$, with ν_j as in Lemma 35,*

$$\text{lo}_{t+1}(y) = \text{fsum}_\downarrow((c_s, \text{lo}_t(y)), (c_m, \text{lo}_t(\nu_j(y)))_{j \leq 2d}; D),$$

$$\text{hi}_{t+1}(y) = \text{fsum}_\uparrow((c_s, \text{hi}_t(y)), (c_m, \text{hi}_t(\nu_j(y)))_{j \leq 2d}; D)$$

(with the value 0 at a). Then $\text{lo}_t \leq v_t \leq \text{hi}_t$ on Ω' for all $t \geq 1$, and $\text{lo}_t(y) = 0 \iff \text{hi}_t(y) = 0 \iff v_t(y) = 0$.

(c) *Let $\underline{\sigma}_t = \text{fsum}_\downarrow((1, \text{lo}_t(y))_{y \sim a}; 1)$ and $\bar{\sigma}_t = \text{fsum}_\uparrow((1, \text{hi}_t(y))_{y \sim a}; 1)$. Then $\underline{\sigma}_t \leq \sigma_t \leq \bar{\sigma}_t$, and both vanish if and only if $\sigma_t = 0$. Hence: $\bar{\sigma}_t < \underline{\sigma}_{t+1}$ implies $f(t) < f(t+1)$; $\bar{\sigma}_{t+1} < \underline{\sigma}_t$ implies $f(t) > f(t+1)$; and $\bar{\sigma}_t = \bar{\sigma}_{t+1} = 0$ implies $f(t) = f(t+1) = 0$.*

(d) *Two elements $m 2^e, m' 2^{e'}$ of $\mathbb{F} \setminus \{0\}$ satisfy $m 2^e < m' 2^{e'}$ if and only if $(e, m) < (e', m')$ lexicographically.*

Proof. (a) If no term has $c_j x_j \neq 0$ the routine returns $0 = s$. Otherwise let E be the largest exponent among the non-zero terms $x_j = m_j 2^{e_j}$ with $c_j \geq 1$, and $s_j = E - e_j \geq 0$. The routine forms $\text{acc} = \sum_j c_j v_j$ with $v_j = \lfloor m_j / 2^{s_j} \rfloor$ (down) or $\lceil m_j / 2^{s_j} \rceil$ (up); for $s_j \geq 63$ these are 0 and 1, because $0 < m_j < 2^{63}$ (the program obtains them by a shift for $s_j \leq 63$ and sets them directly for $s_j \geq 64$, where a shift of a 64-bit word is undefined in C). So $\text{acc} 2^E \leq \sum_j c_j x_j$, resp. $\geq$. A term with $s_j = 0$ exists and contributes $c_j m_j \geq 2^{62}$, and $v_j \leq 2^{63}$, so $2^{62} \leq \text{acc} \leq 15 \cdot 2^{63} < 2^{67}$. Next $\text{num} = 2^{58} \text{acc} < 2^{125}$ and $Q = \lfloor \text{num} / D' \rfloor$, resp. $\lceil \text{num} / D' \rceil = \lfloor (\text{num} + D' - 1) / D' \rfloor$, computed by *divsmall* (Lemma 36(b); $\text{num} + D' - 1 < 2^{126}$); thus $Q 2^{E-58} \leq s$, resp. $\geq s$, and, since $2^{120} \leq \text{num} < 2^{125}$ and $1 \leq D' \leq 15$, $2^{116} < \lfloor 2^{120} / 15 \rfloor \leq Q < 2^{125}$ in both modes. Let $\ell \in [117, 125]$ be the number of binary digits of Q and $\varsigma = \ell - 63 \in [54, 62]$. The mantissa is $m = \lfloor Q / 2^\varsigma \rfloor$, resp. $\lceil Q / 2^\varsigma \rceil$, so $2^{62} \leq m \leq 2^{63}$, and the result $m 2^{E-58+\varsigma}$ is $\leq s$, resp. $\geq s$; if $m = 2^{63}$ (possible only in mode “up”) the pair (m, ς) is replaced by $(2^{62}, \varsigma + 1)$, which represents the same number. The exponent is $E - 58 + \varsigma \in [E - 4, E + 5]$. The result is non-zero, as is s .

(b) Induction on t : $v_{t+1}(y) = \frac{1}{D} [c_s v_t(y) + c_m \sum_j v_t(\nu_j(y))]$ by Lemma 35(b), all coefficients are non-negative, so (a) gives $\text{lo}_{t+1}(y) \leq \frac{1}{D} [c_s \text{lo}_t(y) + \dots] \leq v_{t+1}(y) \leq \frac{1}{D} [c_s \text{hi}_t(y) + \dots] \leq \text{hi}_{t+1}(y)$. By (a), $\text{lo}_{t+1}(y) = 0$ iff every term with a positive coefficient vanishes, iff (induction) the same holds for v_t , iff $v_{t+1}(y) = 0$; likewise for hi . (The coefficients sum to $D \leq 15$, and $1 = 2^{62} 2^{-62} \in \mathbb{F}$.) (c) follows from (a), (b) and $f = \frac{q}{2d} \sigma$. (d) $2^{62} \leq m, m' < 2^{63}$: if $e < e'$ then $m 2^e < 2^{63+e} \leq 2^{62+e'} \leq m' 2^{e'}$. $\square$

In the symmetry-reduced program the sums over $y \sim a$ in (c) and in $F_t^\pm$ are replaced by d times the value at the representative $(N - 2, N - 1, \dots, N - 1)$ (Lemma 35(d)); by the same lemma the reduced program computes the restrictions to $\Omega_\leq$ of exactly the vectors defined above. The exponents are stored in 32-bit signed integers: by (a) they stay in $[-62 - 4t, -62 + 5t]$ after t steps, and the floating phase lasts fewer than 10^5 steps in all runs.

Certificate data of the C engine. For given (d, N, q) the program writes (binary file) the integers $F_t^- = \sum_{y \sim a} L_t(y)$, $F_t^+ = \sum_{y \sim a} U_t(y)$ for $1 \leq t \leq T+1$; the early relations $\text{rel}_t \in \{<, >, =, ?\}$ of Lemma 37(c) for the times $t < t_{\text{sw}}$, where t_{sw} is the first time with $F_{t_{\text{sw}}}^- \geq 2^{48}$ (and “?” afterwards); the tail time T , the first t with $U_{t+1}(y) < L_t(y)$ at every stored non-target site; and the two vectors L_T, U_{T+1} on the stored sites. The post-processor (Python integers) sets rel_t to “<” if $F_t^+ < F_{t+1}^-$, to “>” if $F_t^- > F_{t+1}^+$, and to rel_t otherwise (a contradiction between two decided relations would raise an error; none occurred); it re-checks $U_{T+1} < L_T$ at every stored non-target site and that F_T^-, F_{T+1}^+ are the sums of these vectors over the neighbours of a ; it computes the largest ratio $\gamma = \max_y U_{T+1}(y)/L_T(y) < 1$ exactly; and it forms the verdicts with the same code as the generator of Section 3 (`fpcore.Checker.finish`).

Proposition 38 (soundness of the C certificates). *The data just described satisfy the hypotheses of Proposition 11: $F_t^- \leq 2^P \sigma_t \leq F_t^+$ for $1 \leq t \leq T+1$; every rel_t is “?” or a true relation between $f(t)$ and $f(t+1)$; $v_{T+1} < v_T$ on Ω' , and $v_{T+1} \leq \gamma v_T$. Moreover $F_t^+ > 0$ if and only if $\sigma_t > 0$, so the first t with $F_t^+ > 0$ is $t_0 = d(N-1)$.*

Proof. Enclosures: Lemma 7 with $t_s = 1$, computed without wrap-around by Lemma 36(c), on the representatives by Lemma 35. Relations: by the enclosures, resp. by Lemma 37(c),(d). Tail: $2^P v_{T+1} \leq U_{T+1} < L_T \leq 2^P v_T$ on $\Omega_{\leq} \setminus \{a\}$, hence on Ω' by Lemma 35(d); and $2^P v_{T+1} \leq U_{T+1} \leq \gamma L_T \leq \gamma 2^P v_T$. Supports: $\lceil W/D \rceil = 0$ iff $W = 0$, so by induction $U_t(y) = 0$ iff $v_t(y) = 0$ (as in Lemma 37(b)); the last claim is Lemma 34. $\square$

8.2 Validation of the engine

The following tests are evidence that the compiled program does what Lemmas 36–37 say; they are not part of the proofs.

- (V1) *Byte-identical output of a separately written Python-integer mirror.* `scripts/c128/c128_ref.py` implements the mathematical description above with Python integers and dictionaries of coordinates (no limbs, no index arrays). For 286 small cases ($d = 1$, $N \leq 40$ and $N = 50, 64, 80$; $d = 2$, $N \leq 16$ and $N = 20, 24$; $d = 3$, $N \leq 8$ and $N = 10$; $q \in \{4/5, 1/2, 1\}$ and some cases with $q \in \{2/3, 3/4, 1/3, 3/5\}$; with and without symmetry reduction) and 10 medium cases (up to 74451 time steps, $d = 1$, $N = 200$, $q = 1$, or 1830 stored sites, $d = 2$, $N = 60$) the complete output files (all $F_t^\pm$, all early relations, T , t_{sw} and the final vectors) of the C program are byte-identical with the mirror’s (`certs/c128_refcheck.json`, `c128_refcheck_medium.json`). In the small cases this holds for the optimised build and for a build with the options `-O1 -DCHECK` and `-fsanitize=undefined,address`, in which every division is compared with the native 128-bit division and every range assumption of Lemma 36 is asserted.
- (V2) *Regression on Theorems 12 and 13.* For all 1539 triples (d, q, N) of Table 1 the C certificate has the same t_0, t^*, T , the same verdicts (mode certified, strictly unimodal), the same number of certified strict local maxima and of undecided relations as the certificate of the generator, and the enclosures of $2^P \sigma_t$ at $t^* - 1, t^*, t^* + 1$ intersect (number of disagreements: 0). Since every one of those certificates was confirmed with Python integers by `verify_packed.py`, the C engine agrees with a Python-integer computation on 1539 cases with up to 4×10^5 time steps.
- (V3) *Cross-checks inside the extended ranges* (Table 6): 28 of the new certificates were re-derived by the generator `fp limb.py` (exact Python-integer early phase, then int64 limbs) and 15 by `verify_packed.py` (Python integers only), each a single computation of less than twenty minutes; all have identical t_0, t^*, T and verdicts and intersecting enclosures.

- (V4) *Every certificate computed twice.* Each of the 3824 certificates was computed a second time, in a separate run (`scripts/c128/c128_recheck.py`); the sha256 digest of the complete output file (all $F_t^\pm$, the early relations, T , t_{sw} and the vectors L_T , U_{T+1}) agrees with that of the first run in every case (0 failures). This guards against transient hardware faults, the computer having no error-correcting memory.
- (V5) *Conservation of probability.* Since $\sum_{y \in \Omega'} v_t(y) + \sum_{s < t} f(s) = 1$ and $f = \frac{c_m}{D} \sigma$, the output of a correct run must satisfy

$$D \sum_y \mu(y) L_T(y) + c_m \sum_{s=1}^{T-1} F_s^- \leq D 2^P \leq D \sum_y \mu(y) U_{T+1}(y) + c_m \sum_{s=1}^T F_s^+,$$

where the sums over y run over the stored sites and $\mu(y)$ is the number of lattice sites represented by y . Both inequalities were verified with Python integers for all 3824 certificates; the two sides differ from $D 2^P$ by less than 6.3×10^{-23} in relative terms. This single test ties the whole sequence of $F_t^\pm$ to the final vectors: a wrap-around, a wrong neighbour index or a corrupted entry anywhere in the run would show up as a violation unless it were smaller than this slack.

- (V6) *Reduced versus full lattice.* By Lemma 35 the reduced and the unreduced program must produce the same integers $F_t^\pm$, the same early relations and the same T ; this was checked in every case of (V1) (both variants agree with the mirror) and is implied by (V2), the programs of Section 3 working on the full lattice.

8.3 Results on the extended ranges

d	q	certified N (C engine)	number	new	re-derived by fplimb	by verify_packed	time (s)
1	4/5	2–1500, 2000	1500	899	650, 700, 750, 850, 900, 950, 1200, 1500	700, 900	5310
1	1/2	2–1000	999	700	500, 900	400, 700	1343
1	1	2–400	399	200	300	300	175
2	4/5	2–301, 350, 401, 450, 500, 600	305	144	210, 220, 230, 250	170, 190	9351
2	1/2	2–200	199	120	120, 190	100, 170	1228
2	1	2–160, 180, 200	161	82	120, 136, 149	86, 92, 136, 149	275
3	4/5	2–120, 130, 140, 150, 160	123	64	62, 64, 66, 70, 75	–	9986
3	1/2	2–80	79	48	40, 64	36	865
3	1	2–60	59	28	40	40	90

Table 6: Ranges of N covered by the certificates of the C engine (`certs/c128/*.jsonl`). “New”: the number of values of N not covered by Table 1. Columns 6 and 7: the new values of N whose certificates were re-derived by the programs of Section 3 (identical t_0 , t^* , T and verdicts, intersecting enclosures). Time: total wall-clock time in seconds of the C engine and of the Python post-processing, on a heavily loaded machine.

For $d \in \{1, 2, 3\}$ and $q \in \{4/5, 1/2, 1\}$ let $\mathcal{N}_C(d, q) \supset \mathcal{N}(d, q)$ be the set of N in the third column of Table 6:

$q = 4/5$: $d = 1$: $2 \leq N \leq 1500$ and $N \in \{2000\}$; $d = 2$: $2 \leq N \leq 301$ and $N \in \{350, 401, 450, 500, 600\}$; $d = 3$: $2 \leq N \leq 120$ and $N \in \{130, 140, 150, 160\}$.

$q = 1/2$: $d = 1$: $2 \leq N \leq 1000$; $d = 2$: $2 \leq N \leq 200$; $d = 3$: $2 \leq N \leq 80$.

$q = 1$: $d = 1$: $2 \leq N \leq 400$; $d = 2$: $2 \leq N \leq 160$ and $N \in \{180, 200\}$; $d = 3$: $2 \leq N \leq 60$.

Theorem 39 (modes and unimodality on the extended ranges; computer-assisted, trusted base (T5)). *Let $d \in \{1, 2, 3\}$, $q \in \{4/5, 1/2\}$ and $N \in \mathcal{N}_C(d, q)$, and suppose that (d, q, N) is not one of the three exceptional triples of Theorem 12.*

- (a) *f is strictly unimodal in the sense of Definition 3: $f(t) = 0$ for $t < d(N-1)$, $f(t) < f(t+1)$ for $d(N-1) - 1 \leq t < t^*$ and $f(t) > f(t+1)$ for all $t \geq t^*$, where $t^* = t^*(d, N, q)$ is the integer in the field `mode` of line N of `certs/c128_dd_q4-5.jsonl` (resp. `..._q1-2.jsonl`). Table 7 lists a selection, Appendix A all new values for $d = 2, 3$, and Theorem 45 a formula for $d = 1$.*
- (b) *$v_{t^*+1} < v_{t^*}$ on Ω' (the tail time is $T = t^*$), and $f(t^* + s) \leq \gamma^s f(t^*)$ for all $s \geq 0$ with a number $\gamma = \gamma(d, N, q) < 1$ such that $1 - \gamma \geq$ the field `one_minus_gamma_lower`; over the ranges, $1 - \gamma \geq 1.28 \times 10^{-15}$, 2.22×10^{-14} , 2.48×10^{-13} for $d = 1, 2, 3$ at $q = 4/5$, and $1 - \gamma \geq 9.45 \times 10^{-14}$, 1.27×10^{-13} , 4.18×10^{-12} at $q = 1/2$.*

For the three exceptional triples the C certificate reports exactly one undecided relation, in agreement with the exact ties of Theorem 12(d). No further exceptional case occurs in $\mathcal{N}_C(d, q)$.

Proof. For each (d, q, N) the certificate was produced by `fp128` and `c128_post.py`. By Proposition 38 its data satisfy the hypotheses of Proposition 11, whose parts (i)–(iii) give (a) and (b) from the recorded verdicts (mode certified, strictly unimodal, no undecided relation, $T = t^*$); the first passage time $d(N-1)$ is Lemma 34. The script `scripts/c128/audit_ext.py` re-checks every assertion against the certificate files (`certs/AUDIT_EXT.json`). $\square$

Theorem 40 (modes and multimodality for $q = 1$ on the extended ranges; computer-assisted, trusted base (T5)). *Let $q = 1$, $d \in \{1, 2, 3\}$ and $N \in \mathcal{N}_C(d, 1)$. The smallest maximiser of f is the integer in the field `mode` of `certs/c128_dd_q1-1.jsonl`, and:*

- $d = 1$: *the maximiser is unique except for $N \in \{5, 6, 8, 9\}$, where the maximum is attained at exactly two times ($N = 5$: $t \in \{4, 6\}$; $N = 6$: $t \in \{7, 9\}$; $N = 8$: $t \in \{15, 17\}$; $N = 9$: $t \in \{20, 22\}$; exact arithmetic, Theorem 13); f is strictly unimodal for $N \in \{2\}$; f has at least two strict local maxima, hence is not unimodal, for $N \in \{4-400\}$ (the largest certified number of strict local maxima is 164054); for $N = 3$, f is unimodal in the weak sense but not strictly: $f(3) = f(4)$; equal consecutive non-zero values occur only for $N = 3$ ($f(3) = f(4)$) and $N = 4$ ($f(8) = f(9)$) (exact arithmetic, Theorem 13): for every other pair (N, t) with N in the range and $t \geq d(N-1) - 1$, $f(t) \neq f(t+1)$; the tail time T exceeds t^* for 398 of the 399 values of N (largest ratio $T/t^* = 6.19$).*
- $d = 2$: *the maximiser is unique except for $N \in \{2\}$, where the maximum is attained at exactly two times ($N = 2$: $t \in \{2, 3\}$; exact arithmetic, Theorem 13); f is strictly unimodal for $N \in \{3-8, 10, 12, 14, 16, 19-20, 22-23, 25, 27-37, 39-50, 52-60, 62-85, 87-91, 93-135, 137-148, 150-160, 180, 200\}$; f has at least two strict local maxima, hence is not unimodal, for $N \in \{9, 11, 13, 15, 17-18, 21, 24, 26, 38, 51, 61, 86, 92, 136, 149\}$ (the largest certified number of strict local maxima is 2); for $N = 2$, f is unimodal in the weak sense but not strictly: $f(2) = f(3)$; equal consecutive non-zero values occur only for $N = 2$ ($f(2) = f(3)$) (exact arithmetic, Theorem 13): for every other pair (N, t) with N in the range and $t \geq d(N-1) - 1$, $f(t) \neq f(t+1)$; the tail time T exceeds t^* for 71 of the 161 values of N (largest ratio $T/t^* = 1.50$).*
- $d = 3$: *the maximiser is unique for every N of the range; f is strictly unimodal for $N \in \{2-60\}$; $f(t) \neq f(t+1)$ for every N of the range and every $t \geq d(N-1) - 1$.*

Proof. Propositions 38 and 11(i)–(iv). Ties are not decided by the C engine; the cases with a tie are those of Theorem 13, settled there in exact arithmetic. The four two-dimensional boxes with

two strict local maxima that are not in Theorem 13, $N = 86, 92, 136, 149$, were re-derived with Python integers only (`verify_packed.py`, Table 6), so these four statements do not depend on (T5). $\square$

d	N	$t^* (q = 4/5)$	$\tau_N = q \mathbb{E}T$	$t^*/\mathbb{E}T$	t_{cf}	$(t_{\text{cf}} - t^*)/t^*$	$t^* (q = 1/2)$	$t^* (q = 1)$
1	600	149727	359400.0000	0.333282	173420.722	+0.15825	239564	—
1	700	203844	489300.0000	0.333283	236078.917	+0.15814	326151	—
1	800	266294	639200.0000	0.333284	308381.324	+0.15805	426070	—
1	900	337075	809100.0000	0.333284	390327.944	+0.15799	539320	—
1	1000	416188	999000.0000	0.333284	481918.777	+0.15794	665901	—
1	1200	599411	1438800.0000	0.333284	694033.080	+0.15786	—	—
1	1500	936737	2248500.0000	0.333284	1084536.129	+0.15778	—	—
1	2000	1665588	3998000.0000	0.333284	1928258.797	+0.15770	—	—
2	160	58239	334809.1538	0.139157	57675.051	−0.00968	93183	46590
2	180	74192	433460.4919	0.136930	73475.274	−0.00966	118708	59354
2	200	92116	545868.2190	0.135001	91228.178	−0.00964	147386	73692
2	220	112018	672247.3736	0.133306	110941.295	−0.00961	—	—
2	250	145594	888433.1920	0.131102	144200.860	−0.00957	—	—
2	280	183655	1137075.8483	0.129212	181905.785	−0.00952	—	—
2	300	211529	1321128.5853	0.128090	209520.706	−0.00949	—	—
2	350	289997	1846288.9361	0.125656	287265.236	−0.00942	—	—
2	401	383013	2479252.0945	0.123590	379430.733	−0.00935	—	—
2	450	484778	3181621.6066	0.121895	480274.045	−0.00929	—	—
2	500	601189	3995002.3721	0.120388	595639.350	−0.00923	—	—
2	600	872259	5919943.2731	0.117874	864299.343	−0.00913	—	—
3	60	21175	919226.7397	0.018429	21178.048	+0.00014	33882	16940
3	70	29447	1462871.7509	0.016104	29449.642	+0.00009	47116	—
3	80	39161	2187198.6384	0.014324	39164.223	+0.00008	62659	—
3	90	50339	3118130.1632	0.012915	50342.246	+0.00006	—	—
3	100	62998	4281589.0863	0.011771	63001.731	+0.00006	—	—
3	110	77155	5703498.1687	0.010822	77158.783	+0.00005	—	—
3	120	92824	7409780.1714	0.010022	92827.959	+0.00004	—	—
3	130	110018	9426357.8555	0.009337	110022.540	+0.00004	—	—
3	140	128750	11779153.9820	0.008744	128754.741	+0.00004	—	—
3	150	149031	14494091.3118	0.008226	149035.865	+0.00003	—	—
3	160	170871	17597092.6061	0.007768	170876.432	+0.00003	—	—

Table 7: Selected certified modes of the extended ranges (conventions of Tables 2–4; the rows $N = 600, 160, 60$ repeat the last rows of those tables).

Remark 41 (margins, and agreement with the floating-point values). In the C certificates with a unique maximiser the gap between the lower bound for $2^P \sigma_{t^*}$ and the upper bounds at $t^* \pm 1$ exceeds the width of the enclosure by a factor of at least 10^9 ; the width $F_{t^*}^+ - F_{t^*}^-$ never exceeds 222108 units of 2^{-112} . The longest run has $T = 1665588$ time steps and the largest reduced lattice 695520 sites; the floating phase never lasts more than 69335 steps. The certified modes coincide with all 97 corner-to-corner values of the computations of Sections 5–6 of the article for $q \in \{4/5, 1/2, 1\}$ that fall in the sets $\mathcal{N}_C$ (0 disagreements), including its largest cases $d = 1$, $N = 1000$; $d = 2$, $N = 401$; $d = 3$, $N = 100$.

Corollary 42 (the mode scales like $1/q$ up to one time step; extended). *With $\Delta_N = \frac{4}{5} t^*(\frac{4}{5}) - \frac{1}{2} t^*(\frac{1}{2})$ as in Corollary 18: $-0.7 \leq \Delta_N \leq 0.5$ for $d = 1$, $2 \leq N \leq 1000$; $-0.9 \leq \Delta_N \leq 0.3$ for $d = 2$, $2 \leq N \leq 200$; $-1.2 \leq \Delta_N \leq 0.4$ for $d = 3$, $2 \leq N \leq 80$; all six bounds are attained.*

Proof. Exact arithmetic on the integers of Theorem 39 (`summarise_ext.py`, re-checked by `audit_ext.py`). $\square$

Theorem 43 (accuracy of the closed form on the extended ranges; computer-assisted, trusted bases (T3), (T5)). *Let t^* be the certified mode of Theorem 39 and t_{cf} as in Definition 20.*

- (a) $d = 2$: for $q = 4/5$ and all $10 \leq N \leq 301$ and $N \in \{350, 401, 450, 500, 600\}$: $|t^* - t_{\text{cf}}| \leq 0.00970 t^*$ (worst case $N = 137$); for $q = 1/2$ and all $10 \leq N \leq 200$: $|t^* - t_{\text{cf}}| \leq 0.00971 t^*$ (worst case $N = 133$).
- (b) $d = 3$: for $q = 4/5$ and all $10 \leq N \leq 120$ and $N \in \{130, 140, 150, 160\}$: $|t^* - t_{\text{cf}}| \leq 0.00275 t^*$; for $q = 1/2$ and all $10 \leq N \leq 80$: $|t^* - t_{\text{cf}}| \leq 0.00157 t^*$ (worst case $N = 10$, resp. $N = 10$).
- (c) $d = 1$: for $q = 4/5$ and all $10 \leq N \leq 1500$ and $N \in \{2000\}$: $0.1577 t^* \leq t_{\text{cf}} - t^* \leq 0.2251 t^*$; for $q = 1/2$ and all $10 \leq N \leq 1000$: $0.1579 t^* \leq t_{\text{cf}} - t^* \leq 0.2282 t^*$.

Bounds on sub-ranges are given in Table 8; the enclosure of the relative error for every single N is in `certs/closedform_c128_dd_qq.jsonl`.

Proof. As for Theorem 21 (`scripts/closed_form.py c128`: Arb at 256 bits from the rational enclosures of τ_N of Lemma 19, which were computed for every N of the sets $\mathcal{N}_C(d, 4/5)$; all have relative width below 10^{-28}). Every bound of the statement and every row of Table 8 was re-evaluated with `mpmath.iv` for every N of its range by `audit_ext.py`. $\square$

Remark 44. The extension confirms the picture of Remark 22 with certified numbers (all taken from Table 8). In two dimensions the relative error is negative for every $N \geq 14$ of the ranges (both values of q); its modulus has a flat maximum, 0.00970 at $N = 137$ for $q = 4/5$, and decreases slowly afterwards, apart from fluctuations of order $1/t^*$ caused by the integer rounding of the mode (rows $160 \leq N \leq 301$ and $200 \leq N \leq 301$ and the isolated values of N in the table). In three dimensions the error is positive for every $N \geq 3$ of the ranges at $q = 4/5$ and for every $N \neq 3$ at $q = 1/2$; it falls to 3.17888×10^{-5} (rounded) at $N = 160$ ($q = 4/5$), though not monotonically, for the same reason. In one dimension $t_{\text{cf}}/t^* - 1$ lies in $[0.157704, 0.157705]$ at $N = 2000$; the value suggested by the continuum limit ($X_N \rightarrow \pi^2/2$ and $t^*/\mathbb{E}T \rightarrow \kappa_1$, neither of which is proved in this note) is $\ln(\pi^2)/((\pi^2/2 + 1)\kappa_1) - 1 = 0.157475 \dots$. The closed form is not an approximation of the one-dimensional mode.

Theorem 45 (the one-dimensional mode on the extended range; computer-assisted, trusted bases (T3), (T5)). *Let $d = 1$, κ_1 as in Theorem 23 and $y_N(q) = \kappa_1(N - \frac{1}{2})^2/q$.*

- (a) For $q = 4/5$ and every $10 \leq N \leq 1500$ and $N \in \{2000\}$: $-1.0989 < t^* - y_N < -0.0997$, hence $t^* = \lfloor y_N - 0.0997 \rfloor$.
- (b) For $q = 1/2$ and every $10 \leq N \leq 1000$: $-1.160 < t^* - y_N < -0.164$, hence $t^* = \lfloor y_N - 0.164 \rfloor$.
- (c) On the contiguous ranges, $|t^*/\mathbb{E}T - \kappa_1| \leq c/(N(N - 1))$ with $c = 0.80$ for $q = 4/5$ and $c = 0.50$ for $q = 1/2$.

Proof. Exact rational arithmetic on the integers of Theorem 39 and the rational enclosure of κ_1 (`closed_form.py c128`, `summarise_ext.py`; re-checked for every N , with both endpoints of the enclosure of κ_1 , by `audit_ext.py`). $\square$

9 Conjectures, observations and remaining gaps

Conjecture 46 (unimodality for all N). For $d \in \{1, 2, 3\}$, $q \in \{4/5, 1/2\}$ and every $N \geq 2$ the corner-to-corner first-passage PMF is unimodal, and strictly unimodal except in the three cases of Theorem 12(d).

Status: proved in the weak sense for $d = 1$, $q \leq 1/2$ and all N (Theorem 28); a theorem for the finite ranges of Theorems 12 and 39; *false* for $q = 1$ (Theorem 13), so some restriction on q is necessary. No proof is known to us for $d \geq 2$, nor for $d = 1$ with $1/2 < q < 1$.

d	q	range of N	bound on $ t_{\text{cf}} - t^* /t^*$	worst N	interval containing $(t_{\text{cf}} - t^*)/t^*$
1	1/2	$10 \leq N \leq 1000$	0.228187	10	[0.157934, 0.228187]
1	1/2	$20 \leq N \leq 1000$	0.184401	21	[0.157934, 0.184401]
1	1/2	$35 \leq N \leq 1000$	0.171773	36	[0.157934, 0.171773]
1	1/2	$50 \leq N \leq 1000$	0.167141	51	[0.157934, 0.167141]
1	1/2	$60 \leq N \leq 1000$	0.165552	60	[0.157934, 0.165552]
1	1/2	$100 \leq N \leq 1000$	0.162171	102	[0.157934, 0.162171]
1	1/2	$160 \leq N \leq 1000$	0.160388	161	[0.157934, 0.160388]
1	1/2	$200 \leq N \leq 1000$	0.159792	200	[0.157934, 0.159792]
1	1/2	$300 \leq N \leq 1000$	0.159009	300	[0.157934, 0.159009]
1	1/2	$600 \leq N \leq 1000$	0.158243	600	[0.157934, 0.158243]
1	4/5	$10 \leq N \leq 1500$	0.225045	11	[0.157780, 0.225045]
1	4/5	$20 \leq N \leq 1500$	0.186954	21	[0.157780, 0.186954]
1	4/5	$35 \leq N \leq 1500$	0.172733	35	[0.157780, 0.172733]
1	4/5	$50 \leq N \leq 1500$	0.167595	50	[0.157780, 0.167595]
1	4/5	$60 \leq N \leq 1500$	0.165848	60	[0.157780, 0.165848]
1	4/5	$100 \leq N \leq 1500$	0.162246	101	[0.157780, 0.162246]
1	4/5	$160 \leq N \leq 1500$	0.160415	161	[0.157780, 0.160415]
1	4/5	$200 \leq N \leq 1500$	0.159836	200	[0.157780, 0.159836]
1	4/5	$300 \leq N \leq 1500$	0.159020	300	[0.157780, 0.159020]
1	4/5	$600 \leq N \leq 1500$	0.158247	600	[0.157780, 0.158247]
1	4/5	$N = 2000$	0.157705	2000	[0.157704, 0.157705]
2	1/2	$10 \leq N \leq 200$	0.009704	133	[-0.009704, 0.006710]
2	1/2	$20 \leq N \leq 200$	0.009704	133	[-0.009704, -0.004819]
2	1/2	$35 \leq N \leq 200$	0.009704	133	[-0.009704, -0.007947]
2	1/2	$50 \leq N \leq 200$	0.009704	133	[-0.009704, -0.008933]
2	1/2	$60 \leq N \leq 200$	0.009704	133	[-0.009704, -0.009238]
2	1/2	$100 \leq N \leq 200$	0.009704	133	[-0.009704, -0.009634]
2	1/2	$160 \leq N \leq 200$	0.009690	160	[-0.009690, -0.009640]
2	4/5	$10 \leq N \leq 301$	0.009698	137	[-0.009698, 0.005978]
2	4/5	$20 \leq N \leq 301$	0.009698	137	[-0.009698, -0.004006]
2	4/5	$35 \leq N \leq 301$	0.009698	137	[-0.009698, -0.007897]
2	4/5	$50 \leq N \leq 301$	0.009698	137	[-0.009698, -0.008910]
2	4/5	$60 \leq N \leq 301$	0.009698	137	[-0.009698, -0.009204]
2	4/5	$100 \leq N \leq 301$	0.009698	137	[-0.009698, -0.009491]
2	4/5	$160 \leq N \leq 301$	0.009687	163	[-0.009687, -0.009491]
2	4/5	$200 \leq N \leq 301$	0.009639	200	[-0.009639, -0.009491]
2	4/5	$300 \leq N \leq 301$	0.009495	300	[-0.009495, -0.009491]
2	4/5	$N = 350$	0.009420	350	[-0.009420, -0.009419]
2	4/5	$N = 401$	0.009353	401	[-0.009353, -0.009352]
2	4/5	$N = 450$	0.009291	450	[-0.009291, -0.009290]
2	4/5	$N = 500$	0.009232	500	[-0.009232, -0.009231]
2	4/5	$N = 600$	0.009126	600	[-0.009126, -0.009125]
3	1/2	$10 \leq N \leq 80$	0.001562	10	[0.000052, 0.001562]
3	1/2	$20 \leq N \leq 80$	0.000386	20	[0.000052, 0.000386]
3	1/2	$35 \leq N \leq 80$	0.000182	35	[0.000052, 0.000182]
3	1/2	$50 \leq N \leq 80$	0.000111	51	[0.000052, 0.000111]
3	1/2	$60 \leq N \leq 80$	0.000085	60	[0.000052, 0.000085]
3	4/5	$10 \leq N \leq 120$	0.002749	10	[0.000039, 0.002749]
3	4/5	$20 \leq N \leq 120$	0.000945	21	[0.000039, 0.000945]
3	4/5	$35 \leq N \leq 120$	0.000249	37	[0.000039, 0.000249]
3	4/5	$50 \leq N \leq 120$	0.000195	50	[0.000039, 0.000195]
3	4/5	$60 \leq N \leq 120$	0.000144	60	[0.000039, 0.000144]
3	4/5	$100 \leq N \leq 120$	0.000060	100	[0.000039, 0.000060]
3	4/5	$N = 130$	0.000042	130	[0.000041, 0.000042]
3	4/5	$N = 140$	0.000037	140	[0.000036, 0.000037]
3	4/5	$N = 150$	0.000033	150	[0.000032, 0.000033]
3	4/5	$N = 160$	0.000032	160	[0.000031, 0.000032]

Table 8: Certified bounds for the relative error of the closed form on sub-ranges of the extended ranges (conventions of Table 5).

Conjecture 47 (the target’s neighbours peak last). For $d \in \{1, 2, 3\}$, $q \in \{4/5, 1/2\}$ and every $N \geq 2$, with t^* the largest maximiser of f : $v_{t^*+1} < v_{t^*}$ on Ω' . In words: by the time the flux into the target peaks, the occupation probability of *every* non-target site is already decreasing.

Status: a theorem in the certified ranges (Theorem 12(c),(d), Theorem 39(b)): it holds in all 3205 certificates of the C engine with $q \in \{4/5, 1/2\}$ (3202 non-exceptional ones, where the maximiser is unique and $T = t^*$, and the three exceptional triples, where T is the largest of the exactly known maximisers). With t^* the *smallest* maximiser, $T = t^*$ holds in 3203 of them; the two exceptions are the plateaus $(1, \frac{1}{2}, 3)$ and $(1, \frac{1}{2}, 4)$. By Lemma 5, Conjecture 47 together with monotone increase of f before t^* would give Conjecture 46. For $q = 1$ the statement fails for some N (for example $d = 2$, $N = 4$: $t^* = 16$, $T = 17$).

Conjecture 48 (accuracy of the closed form for all N). For $q \in \{4/5, 1/2\}$ and every $N \geq 10$: $|t^* - t_{\text{cf}}| \leq 0.0100 t^*$ for $d = 2$ and $|t^* - t_{\text{cf}}| \leq 0.0030 t^*$ for $d = 3$.

Status: a theorem, with the sharper constants of Theorems 21 and 43, in the certified ranges ($N \leq 301$ and four larger values in two dimensions, $N \leq 120$ in three, for $q = 4/5$). The evidence beyond them is the continuous-time computations of Sections 5–6 of the article. No proof is known to us; see G1 below.

Numerical observation 49 (log-concavity in $d = 2, 3$). For $d = 2$, $N \leq 7$ and $d = 3$, $N \leq 4$, with $q = 1/2$ and $q = 4/5$, exact rational arithmetic gives $f(t)^2 \geq f(t-1)f(t+1)$ for all t up to four times the tail time (`certs/logconcavity_check.json`). This is an observation on a finite horizon, not a theorem.

Refuted statements.

- “The PMF is unimodal for every $q \in (0, 1]$ ”: false at $q = 1$ (Theorem 13), in one dimension for every $N \geq 4$ (Proposition 32); in one dimension the parity oscillation produces up to 164054 certified strict local maxima ($N = 400$), and for $q = 1$, $d = 1$, $N \in \{5, 6, 8, 9\}$ even the maximum is attained twice; in two dimensions the PMF has two strict local maxima for sixteen values of $N \leq 160$, the four largest being $N = 86, 92, 136, 149$ (Theorem 40).
- “No exact ties occur”: false; $f(3) = f(4)$ for $(d, q, N) = (1, \frac{4}{5}, 4)$ and the maximum is attained twice for $(1, \frac{1}{2}, 3)$ and $(1, \frac{1}{2}, 4)$ (Theorem 12(d)); for $q = 1$, $f(3) = f(4)$ ($d = 1$, $N = 3$), $f(8) = f(9)$ ($d = 1$, $N = 4$) and $f(2) = f(3)$ ($d = 2$, $N = 2$) (Theorems 13, 40).
- “The closed form t_{cf} approximates the mode in every dimension”: false for $d = 1$ (Theorem 21(c)).
- “The PMF is log-concave for every $q < 1$ in one dimension”: false for $q = 4/5$, $N = 3, 4$; for $N = 3$ it is false exactly when $q > 3 - \sqrt{5}$ (Remark 29).

Remaining gaps (what would be needed, in decreasing order of importance).

- G1. *All N* . Theorems 12, 13, 21, 26, 30 and 39–45 are statements about finite sets of N ; the certificates say nothing beyond them. A proof for all N of $|t^* - t_{\text{cf}}| \leq \varepsilon_d t^*$ needs a rigorous version of the two-pole perturbation argument of Section 6 of the article (uniform control of the remainder of the renewal ratio on $[-\mu_2, 0]$ and of the poles $j \geq 2$); this is not done here.
- G2. *Asymptotic laws*. The statements $t^*/\mathbb{E}T \rightarrow \kappa_1$ ($d = 1$), $t^* \sim (4/\pi^2 q)N^2 \ln \ln N$ ($d = 2$) and $t^* \sim (6/\pi^2 q)N^2(\ln N + 3.753)$ ($d = 3$) are not touched by the computations. Theorem 23 proves only that the continuum constant is well defined and gives its value.
- G3. *Identification of C_2* . That C_2 is the limit of $(\tau_N - \frac{8}{\pi}N^2 \ln N)/N^2$ rests on Theorem 4 of the companion note, whose remainder estimate is sketched there. Theorem 26 gives $|(\tau_N - \frac{8}{\pi}N^2 \ln N)/N^2 - C_2| < 0.53213/N^2$ for $2 \leq N \leq 301$ unconditionally.

- G4. *Three-dimensional MFPT constants* $C_3 \approx 4.32046$, $C'_3 \approx -3.887$ (proved in note R2, Theorem 10.1, and stated as Theorem 4.8 of the article) are not certified in this note; rigorous enclosures of τ_N for $d = 3$, $2 \leq N \leq 120$ and $N = 130, 140, 150, 160$, are in `certs/mfpt_d3.jsonl`. (An enclosure of C'_3 is derived in the companion note R2.)
- G5. *Scope*. Only the corner-to-corner geometry, only $q \in \{4/5, 1/2, 1\}$, only discrete time.
- G6. *Ranges*. With Python integers alone (T1) the largest three-dimensional box certified is $N = 60$ (216000 sites, 21175 steps), the limit being computer time: its verification with Python integers took 1794 s of wall-clock time on the loaded machine, and the cost grows roughly like N^5 . With the C engine (T5) the ranges are those of Table 6; they contain all corner-to-corner cases of the discrete-time study of Sections 5–6 of the article. Larger N are a matter of computer time only (the cost per value of N grows like N^3 , N^4 , N^5 in dimensions 1, 2, 3, up to logarithms), the engine being restartable from checkpoints.
- G7. *Trusted base*. Theorems 12 and 13 rest on the correctness of CPython’s integer arithmetic and of the 253-line program `verify_packed.py` (all 1539 certificates), supported by four further programs with which it never disagreed. Theorems 39–45 rest in addition on the C compiler and on the 366-line program `fp128.c` (T5); of the 2285 certificates outside Table 1, 28 were re-derived by the generator and 15 with Python integers only, so for the remaining ones the C program is the only witness (each of them was computed twice with identical output and passes the conservation test (V5) of Section 8). Theorems 21, 23, 25, 26 rest on the correctness of Arb, cross-checked with `mpmath.iv` (and re-evaluated with `mpmath.iv` alone by `audit.py`). No program was formally verified in a proof assistant.

10 Verification matrix, files and reproduction

d	q	certificates	<code>verify_packed</code>	exact	<code>fpcore</code>	time (s): generator, packed, exact		
1	4/5	601	601	159	207	3338	4452	619
1	1/2	299	299	119	119	237	253	305
1	1	199	199	99	99	143	124	362
2	4/5	161	161	49	63	4151	14791	619
2	1/2	79	79	39	39	106	715	533
2	1	79	79	39	39	37	263	80
3	4/5	59	59	19	26	3617	16176	292
3	1/2	31	31	15	15	79	586	158
3	1	31	31	15	15	29	328	17

Table 9: Number of mode certificates and of re-derivations by separately written programs, with wall-clock times in seconds (on a heavily loaded machine; for these single-threaded programs they are upper bounds for the CPU times). “`verify_packed`”: identical verdicts obtained with Python integers only. “exact”: re-derived without any rounding. “`fpcore`”: identical JSON line including the sha256 digest of (L_T, U_{T+1}) .

Separately written programs. All programs were written within this project and run on the same machine; none of the checks below is a verification by a third party. The 1539 certificates were produced by the generator `fp1mb.py`. Of these, 1539 were re-derived by `verify_packed.py` (different data layout, different rounding scheme, different switch time, Python integers only) with identical t_0 , t^* , T and verdicts and with intersecting enclosures; 553 were re-derived in exact arithmetic; 622 were reproduced bit for bit by `fpcore.py`. The number of disagreements between any two programs is 0. The certified modes also agree with the 88 values of the computations of Sections 5–6 of the article that fall in the certified sets.

The C engine (Section 8). 3824 certificates `certs/c128_*.jsonl`, of which 1539 repeat the cases of Table 1 (0 disagreements with the generator’s certificates) and 2285 are new. Of the new

ones, 28 were re-derived by `fplimb.py` and 15 by `verify_packed.py` (0 disagreements; Table 6). The complete output of the C program is byte-identical with that of the Python-integer mirror `c128_ref.py` in 286 small and 10 medium cases, and runs that were interrupted repeatedly (time budget 0.5 s per invocation: 9 interruptions for $d = 1$, $N = 1500$; 4 for $d = 3$, $N = 50$; 2 for $d = 2$, $N = 120$, in the recorded run; all at $q = 4/5$) and continued from their checkpoints produced output files byte-identical with those of the uninterrupted runs (sha256 equal to the field digest of the certificates; `scripts/c128/checkpoint_test.py`). Every certificate was computed twice (3824 second runs, identical sha256 digests, 0 failures; additional wall-clock time 30080 s), and all satisfy the conservation test (V5). Total wall-clock time of all C runs including post-processing: 28623 s (on a heavily loaded machine; for these single-threaded programs this is an upper bound for the CPU time); the longest single certificate took 1501 s in 2 invocation(s) of at most 15 minutes each; the longest cross-check took 938 s.

Sanity checks of the lemmas (`scripts/check_lemmas.py`, output `certs/lemma_checks.json`; none of these is part of a proof). Lemmas 1, 2, 4, 5 (including the geometric bound) were checked in exact rational arithmetic on eleven small cases over a horizon of four tail times; Lemma 7 was checked against exact rationals with $P = 8, 20, 40$; Lemma 8 on 360 random vectors with extreme limb values for all nine (d, q) ; Lemmas 9 and 10 on random vectors against a direct site-by-site implementation and against exact rationals, and Proposition 6 numerically (`scripts/check_packed.py`, output `certs/packed_checks.json`); Lemma 19 against exact rational solves ($q\mathbb{E}T = 8, 27, 416/7, 1175/11, 9368/55$ for $d = 2$, $N = 2, \dots, 6$; 20 and $1161/14$ for $d = 3$, $N = 2, 3$); formula (6.1) and $(q/2)^n = \prod(1 - \theta_j)$ numerically; the series identity (5.1) to 30 digits, and the Step-3 identity, the bound on $|b'_n|$ and the single sign change of φ' of Theorem 23 on a grid (`check_packed.py`); Theorem 28 by `scripts/check_logconcave.py`. The lemmas of Section 8 are checked by `scripts/c128/check_ext_lemmas.py` (output `certs/ext_lemma_checks.json`): Lemma 34 in exact rational arithmetic on 18 small chains; Lemma 35 by comparing the reduced and the full C program in 8 cases (identical $F_t^\pm$, relations, T ; symmetric final vectors); the division identities of Lemma 36 on 6×10^4 random and extreme arguments; the routine `fsum` of Lemma 37(a) on 6×10^4 random inputs against exact rationals (direction of rounding, zero pattern, exponent window; largest relative width 3.1×10^{-18}); Proposition 38 end to end against exact rational arithmetic in 42 small cases (enclosures, early relations, tail time, t_0 , γ ; in all of them the certified tail time is the first exact one); and the closed forms of $f(N - 1)$, $f(N)$, $f(N + 1)$ and of $\min G$ in Proposition 32 against exact iteration of the chain (54 cases) and for all $n < 400$. Further statements are checked by `scripts/check_remarks.py` (output `certs/remark_checks.json`): the identity of Remark 29 for $N = 3$ in exact arithmetic ($t \leq 80$, nine values of q on both sides of $3 - \sqrt{5}$) and the failures of log-concavity for $N = 4$, $q = 4/5$, $t \leq 400$; the continuum value in Remark 44; the counts in the status of Conjecture 47; the junction in Proposition 27; the signs and values quoted in Remarks 22 and 44; and the speed ratios quoted in Section 8. The checkpoint test of the C engine is `scripts/c128/checkpoint_test.py` (output `certs/c128_checkpoint_test.json`).

Files (in the repository: `scripts/` is `code/msc_rigorous_certified/`, `scripts/c128/` is `code/msc_rigorous_certified_c128/`, `certs/` is `data/msc_rigorous_certified/`; shell drivers and logs are not released).

- `scripts/verify_packed.py`: the verifying program for the mode certificates (Python integers only). `scripts/verify_exact.py`: exact rational check for small N . `scripts/fplimb.py`: generator. `scripts/fpcore.py`: reference implementation of the generator's recurrences. `scripts/run_modes.py`, `driver.sh`, `queue.sh`, `run_packed.sh`, `run_packed_range.sh`, `run_packed_exact.sh`: restartable runners.
- `scripts/mfpt_enclosure.py`, `closed_form.py`, `constants.py`, `mfpt2d_remainder.py`: Lemma 19, Theorems 21, 23, 25, 26, 30.
- `scripts/audit.py`: re-checks every numerical assertion of Theorems 12, 13, 21, 26 and 30

against the certificate files (output `certs/AUDIT.json`) and writes the sha256 digests of all certificate files to `certs/MANIFEST.sha256`.

- `scripts/check_lemmas.py`, `check_packed.py`, `check_logconcave.py`, `check_remarks.py`: sanity checks (the last one writes `certs/remark_checks.json`). `scripts/summarise.py`, `tex2md.py`: bookkeeping.
- `scripts/c128/fp128.c`: the C engine (Section 8). `c128_post.py`: certificate from its binary output (Python integers). `c128_run.py`, `regress_tierA.sh`, `extend.sh`, `extend2.sh`, `extend3.sh`, `extend4.sh`: restartable runners. `c128_ref.py`: Python-integer mirror; `c128_validate_ref.py`, `c128_validate_ref2.py`: byte-for-byte comparisons. `xcheck.py`, `xcheck_all.sh`: cross-checks by `fplimb.py` / `verify_packed.py`. `check_ext_lemmas.py`: sanity checks of the lemmas of Section 8. `c128_recheck.py`: second run of every certificate, digest comparison and conservation test (output `certs/c128_recheck*.jsonl`). `checkpoint_test.py`: repeatedly interrupted runs (output `certs/c128_checkpoint_test.json`). `summarise_ext.py`, `audit_ext.py`: bookkeeping and audit of Theorems 39–45 (output `certs/SUMMARY_EXT.json`, `certs/AUDIT_EXT.json`).
- `certs/c128_dd_qq.jsonl`: certificates of the C engine. `certs/xcheck_limb*.jsonl`, `xcheck_packed*.jsonl`: cross-checks. `certs/c128_refcheck.json`, `c128_refcheck_medium.json`, `c128_checkpoint_test.json`: byte-identity tests. `certs/c128_build.json`: compiler, flags and digests of the build. `certs/closedform_c128*.jsonl`, `closedform_summary_c128.json`: Theorem 43.
- `certs/modes_dd_qq.jsonl`: mode certificates (one JSON line per N ; integers as decimal strings). `certs/packed*.jsonl`, `packedexact*.jsonl`, `refcheck*.jsonl`, `exactcheck*.jsonl`: re-derivations. `certs/mfpt_dd.jsonl`: rational enclosures of τ_N . `certs/closedform*.jsonl`, `closedform_summary.json`: Theorem 21. `certs/constants.json`, `mfpt2d_remainder.json`, `lemma_checks.json`, `packed_checks.json`, `logconcavity_check.json`, `SUMMARY.json`, `AUDIT.json`.

Reproduction (PY = `env/venv/bin/python`; every step is deterministic).

- `scripts/driver.sh d qn qd 2 Nmax limb` produces `certs/modes*.jsonl`; PY `scripts/verify_packed.py certs/modes_d2_q4-5.jsonl` verifies a certificate file with Python integers (add `--exact` and an upper limit for N for the exact mode); `scripts/driver.sh d qn qd 2 Nmax ref` re-derives certificates with `fpcore.py`; PY `scripts/verify_exact.py certs/modes_d2_q4-5.jsonl 13` runs the exact rational check.
- PY `scripts/mfpt_enclosure.py d 2 Nmax`; PY `scripts/constants.py`;
PY `scripts/closed_form.py`; PY `scripts/mfpt2d_remainder.py 1500`;
PY `scripts/check_lemmas.py`; PY `scripts/check_packed.py`; PY
`scripts/check_logconcave.py`; PY `scripts/check_remarks.py`; PY
`scripts/summarise.py`; PY `scripts/audit.py`; PY `scripts/tex2md.py`.
- C engine: `cc -O2 -o work/fp128 scripts/c128/fp128.c`; the checking build adds the options `-O1 -g -DCHECK, -fsanitize=undefined,address` and `-fno-sanitize-recover=all` (output `work/fp128_check`); PY `scripts/c128/c128_validate_ref.py`; PY `scripts/c128/c128_validate_ref2.py`; PY `scripts/c128/check_ext_lemmas.py`;
`scripts/c128/regress_tierA.sh`; `scripts/c128/extend.sh` (then `extend2.sh`, `extend3.sh`, `extend4.sh`); `scripts/c128/xcheck_all.sh limb|packed|batch2`;
PY `scripts/c128/c128_recheck.py`; PY `scripts/c128/checkpoint_test.py`;
PY `scripts/closed_form.py c128`; PY `scripts/c128/summarise_ext.py`; PY

`scripts/c128/audit_ext.py`; `scripts/finalise_all.sh` rebuilds every derived file of both tiers and runs both audits. A single certificate: `work/fp128 d N qn qd 1 out.bin` 900 (exit code 75 = checkpoint written, run again), then PY `scripts/c128/c128_post.py out.bin`.

A single certificate takes from milliseconds to 499 s of wall-clock time (generator; longest: $d = 3$, $N = 58$), and its verification with Python integers up to 1794 s ($d = 3$, $N = 60$). The machine was heavily loaded during the runs, so these times and those of Table 9 are upper bounds for the CPU times. The largest state vectors have 216000 sites (Python programs, $d = 3$, $N = 60$) and 695520 stored sites (C engine, $d = 3$, $N = 160$); peak memory use was not recorded.

A All certified modes in two and three dimensions

For completeness, Tables 10–13 list every certified mode $t^*(d, N, q)$ of Theorem 12 for $d = 2, 3$ (read down the columns). In one dimension the modes for $10 \leq N \leq 600$ ($q = 4/5$) and $10 \leq N \leq 300$ ($q = 1/2$) are given by the floor formula of Theorem 30; for $N = 2, \dots, 9$ they are (for $q = 1/2$, $N = 3, 4$: the smaller of the two maximisers) $q = 4/5$: 1, 2, 5, 8, 12, 17, 23, 29; $q = 1/2$: 1, 3, 7, 13, 19, 27, 37, 47

N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*
2	3	23	1030	44	4024	65	9064	86	16189	107	25425	128	36791	149	50301
3	10	24	1127	45	4217	66	9356	87	16581	108	25918	129	37385	150	50999
4	21	25	1228	46	4415	67	9652	88	16977	109	26415	130	37985	151	51701
5	36	26	1334	47	4617	68	9953	89	17379	110	26918	131	38589	152	52407
6	55	27	1445	48	4824	69	10259	90	17785	111	27425	132	39198	153	53119
7	78	28	1560	49	5036	70	10570	91	18196	112	27937	133	39812	154	53836
8	105	29	1679	50	5253	71	10886	92	18612	113	28454	134	40431	155	54557
9	136	30	1804	51	5474	72	11206	93	19032	114	28976	135	41055	156	55284
10	172	31	1932	52	5700	73	11531	94	19457	115	29503	136	41684	157	56015
11	212	32	2066	53	5930	74	11860	95	19888	116	30034	137	42318	158	56752
12	256	33	2203	54	6166	75	12195	96	20323	117	30571	138	42956	159	57493
13	304	34	2346	55	6406	76	12534	97	20762	118	31112	139	43599	160	58239
14	357	35	2493	56	6650	77	12878	98	21207	119	31658	140	44248	180	74192
15	414	36	2645	57	6900	78	13227	99	21656	120	32209	141	44901	200	92116
16	475	37	2801	58	7154	79	13580	100	22110	121	32764	142	45559		
17	541	38	2962	59	7412	80	13939	101	22569	122	33325	143	46222		
18	612	39	3127	60	7676	81	14302	102	23033	123	33891	144	46889		
19	686	40	3297	61	7944	82	14670	103	23502	124	34461	145	47562		
20	765	41	3472	62	8217	83	15042	104	23975	125	35036	146	48240		
21	849	42	3651	63	8494	84	15420	105	24454	126	35616	147	48922		
22	937	43	3835	64	8777	85	15802	106	24937	127	36201	148	49609		

Table 10: $d = 2$, $q = 4/5$: all certified modes.

N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*
2	5	12	410	22	1500	32	3305	42	5842	52	9120	62	13147	72	17930
3	17	13	487	23	1648	33	3526	43	6137	53	9489	63	13592	73	18450
4	34	14	571	24	1803	34	3754	44	6438	54	9866	64	14043	74	18977
5	58	15	663	25	1965	35	3989	45	6747	55	10250	65	14503	75	19512
6	88	16	761	26	2135	36	4232	46	7064	56	10641	66	14969	76	20055
7	125	17	867	27	2312	37	4482	47	7388	57	11040	67	15444	77	20606
8	168	18	979	28	2496	38	4739	48	7720	58	11446	68	15926	78	21164
9	219	19	1098	29	2688	39	5004	49	8059	59	11860	69	16415	79	21729
10	275	20	1225	30	2886	40	5276	50	8405	60	12282	70	16913	80	22303
11	339	21	1359	31	3092	41	5555	51	8759	61	12711	71	17417		

Table 11: $d = 2$, $q = 1/2$: all certified modes.

N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*
2	8	10	422	18	1561	26	3488	34	6240	42	9845	50	14323	58	19692
3	24	11	523	19	1757	27	3786	35	6644	43	10357	51	14945	59	20427
4	49	12	635	20	1967	28	4097	36	7061	44	10882	52	15582	60	21175
5	85	13	760	21	2188	29	4421	37	7491	45	11421	53	16232		
6	131	14	896	22	2423	30	4759	38	7935	46	11974	54	16896		
7	187	15	1044	23	2670	31	5109	39	8392	47	12541	55	17574		
8	254	16	1204	24	2930	32	5473	40	8863	48	13121	56	18266		
9	332	17	1376	25	3202	33	5850	41	9347	49	13715	57	18972		

Table 12: $d = 3$, $q = 4/5$: all certified modes.

N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*
2	12	6	210	10	676	14	1434	18	2498	22	3878	26	5581	30	7615
3	39	7	300	11	838	15	1671	19	2813	23	4273	27	6058	31	8176
4	80	8	408	12	1018	16	1927	20	3148	24	4689	28	6557	32	8758
5	137	9	533	13	1217	17	2203	21	3503	25	5125	29	7075		

Table 13: $d = 3$, $q = 1/2$: all certified modes.

Tables 14–17 list the additional modes of Theorem 39 for $d = 2, 3$ (C engine; the values of N not contained in the tables above).

N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*
161	58990	183	76755	205	96906	226	118375	247	142035	268	167891	289	195950		
162	59746	184	77619	206	97879	227	119452	248	143216	269	169177	290	197341		
163	60507	185	78488	207	98856	228	120534	249	144403	270	170468	291	198738		
164	61272	186	79362	208	99839	229	121621	250	145594	271	171765	292	200139		
165	62043	187	80241	209	100827	230	122713	251	146790	272	173066	293	201545		
166	62818	188	81125	210	101819	231	123810	252	147992	273	174372	294	202956		
167	63599	189	82014	211	102817	232	124911	253	149198	274	175683	295	204372		
168	64384	190	82907	212	103819	233	126018	254	150409	275	176999	296	205794		
169	65175	191	83806	213	104827	234	127130	255	151625	276	178320	297	207220		
170	65970	192	84710	214	105839	235	128247	256	152847	277	179646	298	208651		
171	66770	193	85618	215	106857	236	129368	257	154073	278	180977	299	210087		
172	67575	194	86532	216	107879	237	130495	258	155304	279	182314	300	211529		
173	68385	195	87450	217	108906	238	131626	259	156540	280	183655	301	212975		
174	69200	196	88373	218	109939	239	132763	260	157782	281	185001	350	289997		
175	70019	197	89302	219	110976	240	133905	261	159028	282	186352	401	383013		
176	70844	198	90235	220	112018	241	135051	262	160279	283	187708	450	484778		
177	71674	199	91173	221	113065	242	136203	263	161535	284	189069	500	601189		
178	72508	201	93064	222	114117	243	137359	264	162796	285	190435	600	872259		
179	73348	202	94017	223	115174	244	138521	265	164063	286	191807				
181	75042	203	94975	224	116236	245	139687	266	165334	287	193183				
182	75896	204	95938	225	117303	246	140858	267	166610	288	194564				

Table 14: $d = 2$, $q = 4/5$: the additional certified modes (Theorem 39).

N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*
81	22884	99	34651	117	48914	135	65689	153	84991	171	106832	189	131222
82	23472	100	35377	118	49779	136	66695	154	86138	172	108120	190	132652
83	24068	101	36112	119	50653	137	67709	155	87293	173	109416	191	134090
84	24672	102	36854	120	51535	138	68730	156	88455	174	110720	192	135536
85	25284	103	37604	121	52424	139	69760	157	89625	175	112032	193	136990
86	25903	104	38361	122	53321	140	70797	158	90803	176	113351	194	138451
87	26530	105	39127	123	54225	141	71842	159	91989	177	114679	195	139921
88	27164	106	39900	124	55138	142	72895	160	93183	178	116014	196	141398
89	27806	107	40680	125	56058	143	73955	161	94384	179	117357	197	142883
90	28456	108	41469	126	56986	144	75024	162	95594	180	118708	198	144376
91	29114	109	42265	127	57922	145	76100	163	96811	181	120067	199	145877
92	29779	110	43069	128	58866	146	77184	164	98036	182	121434	200	147386
93	30452	111	43881	129	59817	147	78276	165	99269	183	122809		
94	31132	112	44700	130	60776	148	79375	166	100510	184	124191		
95	31821	113	45527	131	61743	149	80483	167	101759	185	125582		
96	32517	114	46362	132	62718	150	81598	168	103015	186	126980		
97	33220	115	47205	133	63701	151	82722	169	104280	187	128386		
98	33932	116	48055	134	64691	152	83853	170	105552	188	129800		

Table 15: $d = 2$, $q = 1/2$: the additional certified modes (Theorem 39).

N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*
61	21938	71	30353	81	40213	91	51538	101	64346	111	78654	130	110018
62	22715	72	31274	82	41279	92	52752	102	65709	112	80167	140	128750
63	23507	73	32209	83	42360	93	53980	103	67087	113	81696	150	149031
64	24312	74	33158	84	43456	94	55224	104	68480	114	83240	160	170871
65	25132	75	34122	85	44566	95	56482	105	69888	115	84800		
66	25966	76	35101	86	45691	96	57756	106	71312	116	86374		
67	26815	77	36094	87	46831	97	59044	107	72750	117	87964		
68	27678	78	37102	88	47985	98	60347	108	74203	118	89568		
69	28555	79	38124	89	49155	99	61665	109	75671	119	91189		
70	29447	80	39161	90	50339	100	62998	110	77155	120	92824		

Table 16: $d = 3$, $q = 4/5$: the additional certified modes (Theorem 39).

N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*	N	t^*
33	9361	39	13428	45	18275	51	23914	57	30357	63	37612	69	45690
34	9985	40	14182	46	19160	52	24932	58	31509	64	38901	70	47116
35	10631	41	14957	47	20066	53	25972	59	32684	65	40213	71	48566
36	11298	42	15753	48	20995	54	27034	60	33882	66	41548	72	50039
37	11987	43	16572	49	21946	55	28119	61	35103	67	42905	73	51535
38	12697	44	17413	50	22919	56	29227	62	36346	68	44286	74	53055
												80	62659

Table 17: $d = 3$, $q = 1/2$: the additional certified modes (Theorem 39).

References

- [1] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 5th ed., Oxford University Press, 1979, Theorem 357 (Jacobi’s identity for $\prod(1 - x^k)^3$).
- [2] T. M. Apostol, *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed., Springer, 1990, Theorem 3.1 (Dedekind’s functional equation $\eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau)$).
- [3] E. Seneta, *Non-negative Matrices and Markov Chains*, 2nd ed., Springer, 1981, Theorem 1.1 (Perron–Frobenius theorem for primitive matrices).
- [4] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Trans. Comput.* 66 (2017) 1281–1292.
- [5] The proofs of the mean first-passage time formulas of the article: the single-sum formula for the corner-to-corner MFPT on the reflecting square (Theorem 4.2 of the article, proof in its Supplementary Section S3) and its large- N expansion (Theorem 4.5 of the article, proof in its Supplementary Section S4).